

Composition Is Where Obfuscation Robustness Breaks

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Fine-tuning a code model on obfuscated programs raises its accuracy on those programs by roughly nineteen points. We ask what that number is made of: competence at reading the program, or competence at the surface a particular transform leaves behind. We separate the two with tests that share no machinery — whether the gain survives being stacked with a transform family the model has never seen, and whether it survives the task being run backwards. The field’s existing answers, inference-time routing, weight-space merging, and training on a breadth of transforms, fail both — merging outright, routing by matching breadth wherever the two are measured together. Breadth composes on stacks built from what it saw and loses that advantage the moment an unseen family enters, a gap significant on every one of six models across four lineages; and every method that fits the forward task buys its gain by losing backward competence on the very same programs. Three further instruments locate what is learned instead: the surface a transform leaves, which is exactly what an obfuscator is free to change. That diagnosis names its own remedy. We anchor the model to the one thing a semantics-preserving transform leaves invariant, its own behaviour on the unobfuscated parent of each program, through a paired-consistency objective that trains on exactly the rows breadth trains on. Re-running the same two tests, the anchored model is the only one whose advantage *grows* as input diverges from training, rising above the clean-code control on stacks containing an unseen family, and the only one whose forward gain costs nothing backwards. Its repair holds on six of eight models and reverses on one, and on the unseen family alone it is as robust as clean-code tuning and no more. Adaptation to obfuscated code buys recombination of what the data covered; anchoring recovers what that costs; neither buys invariance.

1 Introduction

Code obfuscation is widely used to make software difficult to inspect while preserving its intended functionality. In benign settings, developers employ obfuscation to protect intellectual property, implementation details, and proprietary business logic. In adversarial settings, however, malware authors routinely exploit the same techniques to conceal malicious behavior and frustrate program analysis and reverse engineering. Common transformations include identifier renaming, control-flow restructuring, dead-code insertion, expression rewriting, call indirection, etc. Although such transformations are intended to preserve program semantics, they can substantially obscure how those semantics are expressed in the source code, thereby increasing the difficulty of understanding the behavior of malware code [2, 3, 10, 11].

This difficulty is well established for human analysts. Nguyen *et al.* [7], for example, showed that code obfuscation significantly reduces humans’ accuracy in reasoning about program outputs. Such findings are particularly important for malware analysis, where analysts must often infer the behavior of code whose surface structure has been deliberately engineered to resist comprehension.

Large language models (LLMs) offer a promising alternative for automating such reasoning. Models have demonstrated strong capabilities in code generation, summarization, vulnerability detection, and program understanding [1, 4–6, 8]. However, most evaluations are performed on naturally written programs, where meaningful identifiers, familiar control structures, and common coding idioms provide abundant lexical and structural cues. Obfuscated programs remove or distort many of these cues while retaining the underlying computation, creating a useful stress test of whether an LLM actually reasons about program behavior or instead depends on surface regularities.

Recent studies suggest that this distinction matters. Rong *et al.* [?] showed that semantics-preserving obfuscations can cause substantial degradation in LLM program reasoning, including program-output prediction. Similarly, Nikiema *et al.* [9] reported a statistically significant decline in model performance as the complexity of obfuscation increases. Together, these findings indicate that this limitation is especially problematic for security applications: an AI-assisted malware analyzer should not cease to understand a malicious computation merely due to obfuscation.

An important challenge, however, has received considerably less attention: *real obfuscated programs rarely contain only one transformation*. Obfuscators can combine multiple techniques so that, for example, renamed identifiers are embedded inside flattened control flow, rewritten expressions, and additional irrelevant code. Such *stacked obfuscation* creates a fundamentally harder deployment setting. Success on each transformation independently does not necessarily imply success when those transformations occur together. Consequently, simply demonstrating that a model can be adapted to individual obfuscations is insufficient. For practical malware analysis, the learned capability must also *compose*.

This observation naturally suggests several ways to compose knowledge learned from individual transformations. One could train a specialist for each obfuscation and use a *router* to select the appropriate specialist at inference time. Alternatively, one could combine the specialists in parameter space through *model merging*. A third, and seemingly more direct, strategy is to increase *training breadth* by fine-tuning a single model on examples from many individual obfuscations, hoping that knowledge learned separately will recombine when transformations are stacked. We systematically evaluate these three strategies and find that none provides the desired robustness. The learned router is worth about a point over a random routing policy and nothing over breadth training; merging specialized models performs at or below the clean-code-tuned control; and broader training, although effective on stacks assembled from transformations represented during training, introduces a substantial performance penalty on an unseen obfuscation family. Thus, broader exposure can improve recombination within the training distribution while simultaneously making the model *less* robust outside that distribution.

Our analysis reveals why. The apparent robustness obtained from obfuscation-specific adaptation does not primarily arise from learning the semantics that remain invariant under transformation. Instead, models learn regularities associated with the *surface artifacts* left by particular obfuscators. Evidence for this behavior appears across multiple analyses: adaptation transfers poorly between distinct mechanisms within the same obfuscation family; the benefit of broad training is substantially stronger when recognizable identifier cues remain available; and improvements disappear when the same programs are queried in a different reasoning direction. Thus, stacking exposes a deeper problem than merely insufficient training coverage: obfuscation-specific fine-tuning can teach the model how particular transformations *look* without teaching it to consistently recover the behavior that those transformations preserve.

This diagnosis suggests a different principle for robust adaptation. Rather than attempting to compose transformation-specific knowledge, we anchor learning to something that is invariant across all such transformations: the model’s behavior on the corresponding *unobfuscated program*. For every obfuscated training program x_{obf} , we retain its clean parent x_{clean} . A teacher model, tuned for reasoning over clean code, processes x_{clean} , while the student processes x_{obf} . In addition to the ordinary task loss, we minimize the KL divergence between their output distributions:

$$\mathcal{L} = \text{CE}(y \mid x_{\text{obf}}) + \lambda D_{\text{KL}}(p_{\text{teacher}}(\cdot \mid x_{\text{clean}}) \parallel p_{\text{student}}(\cdot \mid x_{\text{obf}})). \quad (1)$$

The first term exposes the student to obfuscated code, whereas the second explicitly encourages it to preserve the behavioral response of a model reasoning over the semantics-equivalent clean program. Importantly, this objective does not require recognizing which obfuscations are present,

selecting among specialists, merging transformation-specific parameters, or recovering the original source code. Instead, it changes the learning target itself: rather than anchoring adaptation to the surface patterns of individual transformations, it anchors the student to the clean-code behavior that those transformations should preserve.

Our empirical results demonstrate the advantage of this *paired-consistency anchoring*. In contrast to routing, model merging, and breadth-only training, the KL-based objective retains the benefits of learning from diverse obfuscations without paying the corresponding penalty on an unseen obfuscation family. At the 7B scale, the anchored model improves over breadth training by 4.59 percentage points on the unseen family while introducing no degradation on clean code. Additional stress tests show that the resulting model continues to perform well under deeper compositions and transfers to a second programming language. At the same time, our analysis carefully bounds the claim: the method does not create a new ability to solve arbitrary unseen obfuscations beyond that of the clean-code teacher. Rather, it prevents obfuscation adaptation from destroying capabilities the model already possesses. This distinction is important: robust reasoning over obfuscated programs requires not only acquiring adaptation gains, but also preserving semantic competence as the surface form of the program changes.

Contributions. This paper makes the following main contributions:

1. A two-test diagnostic for whether obfuscation adaptation generalises or memorises, and its verdict on the existing composition strategies. We ask whether competence learned on individual semantics-preserving transformations survives leaving the distribution it was fit to, in two independent ways: being stacked with a transformation family the model has never seen, and being asked backwards (given a return value, produce a call that yields it). Inference-time routing, weight-space merging, and broad multi-transformation training all fail both. Breadth’s advantage on stacks of seen transformations vanishes once an unseen family enters — a gap significant on six of six models across four lineages — and every method that fits the forward task buys its gain by losing backward competence on the same programs.

2. An empirical characterisation of what adaptation learns instead. With complementary analyses across transformation mechanisms, identifier dependence, and confidence behaviour, we show that obfuscation adaptation predominantly learns the surface regularities a transformation leaves behind rather than an invariance to semantics-preserving change. Because that surface is exactly what an obfuscator is free to alter, the diagnosis names its own remedy.

3. A clean-behaviour anchoring objective, derived from that diagnosis. We introduce a paired-consistency objective that anchors an obfuscated-code student to a clean-code-tuned teacher reading the unobfuscated parent of the same program — the one thing a semantics-preserving transformation leaves invariant. The objective trains on exactly the rows breadth training uses, so breadth is its data-matched control. Across eight models and four lineages it removes breadth’s unseen-family degradation on six and reverses on one, and its clean-code cost is absent on five; we report both qualifiers rather than the unconditional claim.

4. The same two tests, re-run on the anchored model, with the bound stated. On a new divergence ladder of stacks that contain an unseen transformation family — a stimulus absent from prior work — the anchored model’s advantage *grows* as the input diverges from training, rising above the clean-code control where breadth falls, and it is the only supervised-fine-tuning arm whose forward gain costs nothing backwards. It does not acquire backward competence, and on the unseen family alone it is as robust as clean-code tuning and no more: anchoring repairs what adaptation breaks rather than creating new invariance.

2 Research Questions

Fine-tuning a code model on obfuscated programs raises its accuracy on those programs by roughly nineteen points. Every question below asks what that number is made of. Adaptation can raise forward accuracy by teaching the model to *read* the program better, or by teaching it the surface a particular transform leaves behind; only the first deserves to be called robustness. We separate them with two tests that share no machinery, committed before any method was on the table:

- **Composition.** Does competence on single transforms survive being stacked with a transform family the model has never seen?
- **Reversal.** Does it survive the task being run backwards — given a return value, produce a call that yields it? No adapter is trained on this direction, so any backward gain is transferred program understanding rather than task fitting.

RQ1: Do standard adaptation methods generalise? We apply both tests to the field’s existing strategies — inference-time routing over per-transform specialists, weight-space merging of those specialists, and broad multi-transform fine-tuning — treated as baselines rather than proposals, and then locate what they learn instead, because that is what determines the remedy.

RQ2: Does anchoring to clean-code behaviour repair what adaptation breaks? The diagnosis names its own remedy: anchor the model to its own behaviour on the unobfuscated parent of each program. We derive a paired-consistency objective, isolate its mechanism by ablating the teacher, and measure it across eight models and four lineages.

RQ3: Does the anchored model survive the same tests? We re-run RQ1’s two tests on the anchored model and state the bound: a new capability on unseen obfuscation, or a preservation mechanism that protects the teacher’s competence from surface-level change.

3 Setup

3.1 Models

Table 1 lists the panel: eight instruct models from four lineages (Meta, BigCode, Google, IBM), code-specialised and general, 7B to 34B, plus one pretrained checkpoint used only for the one-shot baseline. Every adapter is a LoRA with the same rank and target modules on every model, and one prompt builder serves training, evaluation and attention extraction.

Table 1. **The model panel.** Four lineages, code-specialised and general instruct models, 7B to 34B. *Chat template: system* accepts a system role; *merged* folds the system text into the user turn; *plain* has no template and is used for the one-shot baseline only. Parameter counts are nominal. Every adapter is a LoRA with the same rank and target modules on every model.

Model	Checkpoint	Lineage	Type	Params (B)	Layers	Hidden	Chat template
CodeLlama-7B	codellama/CodeLlama-7b-Instruct-hf	Meta	coder	6.7	32	4096	system
CodeLlama-13B	codellama/CodeLlama-13b-Instruct-hf	Meta	coder	13.0	40	5120	system
CodeLlama-34B	codellama/CodeLlama-34b-Instruct-hf	Meta	coder	33.7	48	8192	system
Llama-3.1-8B	meta-llama/llama-3.1-8B-Instruct	Meta	instruct	8.0	32	4096	system
StarCoder2-15B	bigcode/starcoder2-15b-instruct-v0.1	BigCode	coder	15.5	40	6144	merged
Gemma-3-12B	google/gemma-3-12b-it	Google	instruct	12.2	48	3840	system
CodeGemma-7B	google/codegemma-7b-it	Google	coder	8.5	28	3072	merged
Granite-3.1-8B	ibm-granite/granite-3.1-8b-instruct	IBM	instruct	8.2	40	4096	system
Llama-3.1-8B (base)	meta-llama/llama-3.1-8B	Meta	pretrained	8.0	32	4096	plain

3.2 Dataset

Table 2 summarises the corpus. The split is by program, never by row; three input cases per program are fixed and their gold outputs executed from the clean parent, so the grader is independent of

any transform. Coverage differs by condition because structural transforms decline some programs by design and the unseen family needs enough encodable sites; every paired contrast is computed on the programs common to the cells it compares, fixed before evaluation.

Table 2. **The dataset.** One Python corpus, split by program so no obfuscated variant of a training program can reach evaluation.

Property	Value
Base programs (Python)	2231: apps 1584, cruxeval 543, humaneval 104
Split	by program, seed 17: train 1563 / val 111 / held-out 557
Input cases per program	3, gold output executed from the clean parent
Program length	mean 8.4 LOC, max 57
Held-out eval, clean	557 programs $\times$ 3 cases = 1671 items
Coverage per condition	Tables 3 and 4; structural transforms decline some programs by design, the unseen family needs ≥ 3 encodable sites
Training rows per adapter	one condition's train split $\times$ 3 cases (4689); breadth pools the five seen conditions
Paired contrasts	on the programs common to every cell in the contrast, fixed before evaluation; program-clustered bootstrap, 2,000 resamples
JavaScript	168 held-out programs (504 items), transferred intact; not regenerable on the cluster
Grading	strict normalised exact match, no containment; the backward task executes the produced call against the clean parent

3.3 Obfuscation taxonomy

Single transforms (Table 3) fall into three families: identifier (L1b, L1r, L2), structural (S1, S2, and S2's two halves S3 and S4) and encoding (X1 and its halves). Each is applied to the clean parent and never stacked during training. H1 is the quarantined held-out obfuscator; its evaluation budget was spent under a pre-registered two-pass discipline, and every unseen-family number in this paper is measured on its trainable sibling X1, which reproduces H1's difficulty at $r = 0.9992$ (mean absolute difference 0.48 points). Stacks (Table 4) are a separate namespace built by applying parts to the parent in order; order matters, and no system is trained on a stack. Ten stacks contain only seen transforms (six at depth two, four at depth three to four); six contain the unseen family or one of its halves and are the stimulus RQ4 turns on.

3.4 Systems

Every system is a LoRA on the same base model, evaluated on still-obfuscated code. *Clean LoRA* (tuned_L0) is tuned on unobfuscated programs and is the control throughout; one *specialist* per seen transform; *breadth* (mono_all) is one adapter trained on all five seen transforms pooled; *family* (tuned_X1) is trained on the unseen family's sibling and is the ceiling for that family. *Routing* mixes the specialists with a learned gate (mole_router), its argmax, a fixed uniform gate or a gate frozen at random initialisation; the last two separate what the mixture is worth from what the routing is worth. *Merging* combines the specialists in weight space (TIES, DARE-TIES, DARE-linear) or anchors the merge on the clean adapter. The *anchored* objective (cons_lam3) is Eq. 1 with $\lambda = 3$, a plateau from a sweep on the trainable grid: the teacher is a frozen clean-code-tuned adapter reading the unobfuscated parent, the student never sees the parent, and the objective trains on exactly the rows breadth trains on (26,841 rows, 1,257 steps for both); at $\lambda = 0$ it reduces to breadth identically, which makes breadth the data-matched control. A one-shot in-context baseline completes the set.

Table 3. **Single transforms.** Each is applied to the clean parent and never stacked. *Cap* is the maximum growth factor over the parent above which a program is declined rather than clipped; *programs / items* is held-out coverage. H1 is the quarantined held-out obfuscator whose evaluation budget is spent; X1 is its trainable sibling and carries every unseen-family number.

Code	Family	Transform	Trainable	Cap	Programs / items	Role
L0	none	normalised original: comments and docstrings stripped, 4-space indent	yes	1.2×	557 / 1671	clean reference
L1b	identifier	adversarial renaming of all bindings incl. the entry function, from semantic-inversion tables (max→min) and an unrelated domain vocabulary	yes	2.0×	553 / 1659	ladder
L1r	identifier	random hex renaming of all bindings (v_a3f2, f_9c1e)	yes	2.0×	557 / 1671	ladder
L2	identifier	sequential minification of all bindings (a, b, ..., aa) plus type-annotation stripping	yes	1.5×	557 / 1671	ladder
S1	structural	control-flow flattening of the entry function into a dispatch loop with randomised state ids and shuffled cases	yes	6.0×	416 / 1248	ladder
S2	structural	opaque predicates (computed always-true/false guards) plus never-called dead helpers	yes	4.0×	556 / 1668	ladder
S3	structural	dead-code insertion only: 1–2 never-called module-level helpers (must be ignored)	yes	4.0×	557 / 1671	stacks only
S4	structural	opaque predicates only: 1–3 computed guards at statement boundaries (must be reasoned about)	yes	4.0×	556 / 1668	stacks only
X1	encoding	every string literal through an inline XOR decoder, every + − ^ through a type-guarded algebraic identity, int literals expanded	yes	12.0×	405 / 1215	unseen family
X1m	encoding	X1’s arithmetic half only (mechanisms: [mba])	yes	12.0×	351 / 1053	mechanism ablation
X1s	encoding	X1’s string half only (mechanisms: [str]); site bar 1	yes	12.0×	246 / 738	mechanism ablation
H1	heldout	string encoding (base64 / RC4 string array) plus guarded mixed-boolean-arithmetic rewriting	never	12.0×	–	quarantined

3.5 Measurement

Forward accuracy is strict normalised exact match on the produced output, with no containment matching; the backward task is graded by executing the produced call against the program as the model was shown it — the obfuscated variant, not the clean parent — and accepting any arguments that return the gold value, which is what the prompt asks for. We also report the stricter rate at which the *gold* arguments are recovered. Every contrast is a paired difference on the program subset common to its cells, with a program-clustered bootstrap of 2,000 resamples; bold intervals exclude zero. Decision rules were pre-registered before submission and are reported by their verdicts even when inconclusive. A cell whose format-failure rate exceeds 0.25 measures the prompt contract

Table 4. **Stacked composites.** A separate namespace from the ladder: parts are applied to the clean parent left to right, and order matters because composition does not commute. Size caps were calibrated on a 40-program build at cap 60 ($p95 \times 1.25$) so that the cap is never the binding constraint. No system is ever trained on a stack; the breadth and family adapters see single transforms only.

Code	Parts, in order	Depth	X1 inside	Cap	Prog./items	Group	Purpose
C_L1b_S1	L1b→S1	2	no	8.0×	418 / 1254	seen d2	does L1b's special-ist advantage survive stacking?
C_L1r_S1	L1r→S1	2	no	8.0×	418 / 1254	seen d2	headline identifier ⊗ structural stack
C_S1_L1r	S1→L1r	2	no	8.0×	418 / 1254	seen d2	order control for C_L1r_S1 (renames S1's state variables)
C_L2_S4	L2→S4	2	no	5.0×	556 / 1668	seen d2	identifier ⊗ the reason-about-it half of S2
C_L1r_S3	L1r→S3	2	no	5.0×	557 / 1671	seen d2	identifier ⊗ the ignore-it half of S2
C_S4_S3	S4→S3	2	no	5.0×	556 / 1668	seen d2	positive control: re-constructs S2 by composition
C3_L1r_S3_S4	L1r→S3→S4	3	no	10.0×	556 / 1668	seen d3–4	depth 3, identifier first
C3_S1_S3_S4	S1→S3→S4	3	no	14.0×	395 / 1185	seen d3–4	depth 3, structural only
C3_L1r_S1_S4	L1r→S1→S4	3	no	14.0×	418 / 1254	seen d3–4	depth 3, identifier first, flattening inside
C4_L1r_S1_S3_S4	L1r→S1→S3→S4	4	no	18.0×	418 / 1254	seen d3–4	depth 4
C_L1r_X1	L1r→X1	2	yes	10.0×	405 / 1215	unseen d2	unseen family last, after renaming
C_X1_S1	X1→S1	2	yes	12.0×	321 / 963	unseen d2	unseen material itself flattened
C_S2_X1	S2→X1	2	yes	22.0×	555 / 1665	unseen d2	unseen family applied to dead code too
C_L1r_X1m	L1r→X1m	2	yes	10.0×	351 / 1053	half-unseen d2	half an unseen family (arithmetic)
C_S1_X1s	S1→X1s	2	yes	14.0×	418 / 1254	half-unseen d2	half an unseen family (strings)
C3_L1r_S1_X1	L1r→S1→X1	3	yes	16.0×	417 / 1251	unseen d3	one unseen component at depth 3

rather than the task and enters no mean; untuned format failure measures adherence to that contract, not capability, so screening base models by it is unsound.

4 Every method against every obfuscation

Tables 5–13 give the raw accuracy of every method on every condition and stack, one table per model, for the eight panel models and the 1.5 B model the project piloted on. Each cell reads *output prediction / input prediction*: the forward task on the left, the backward task on the right. Bold rows average each group of conditions, and an accuracy is coloured where it sits above or below the untuned model in the same direction. Table 5 is the reference the research questions draw

on, because CodeLlama-7B is the one model on which every arm exists. Every number is a strict exact-match accuracy on the held-out programs of that condition; nothing is normalised. On that model the untuned baseline scores 0.257 on clean code, 0.168–0.207 on the single seen transforms, 0.119 on the unseen family and 0.089–0.186 on the stacks; every tuned arm lands between 0.35 and 0.43 on the single seen transforms and their stacks, and the columns separate on the unseen family and on the stacks that contain it. The backward half is graded by execution, and where it is missing the arm was not run in that direction. The routing and merging arms on every model that has them are in Appendix C.

Table 5. **Every method against every obfuscation, CodeLlama-7B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.257 / 0.285	0.302 / -	0.427 / 0.324	0.413 / 0.276	0.422 / 0.297	0.429 / 0.305	0.426 / 0.329
L1b	0.198 / 0.265	0.255 / -	0.358 / 0.251	0.388 / 0.279	0.402 / 0.294	0.381 / 0.300	0.373 / 0.297
L1r	0.207 / 0.315	0.265 / -	0.378 / 0.327	0.387 / 0.276	0.395 / 0.293	0.383 / 0.312	0.381 / 0.352
L2	0.202 / 0.282	0.272 / -	0.382 / 0.324	0.378 / 0.276	0.398 / 0.275	0.386 / 0.328	0.375 / 0.342
S1	0.168 / 0.261	0.216 / -	0.379 / 0.188†	0.389 / 0.262	0.400 / 0.280	0.395 / 0.255	0.376 / 0.308
S2	0.193 / 0.330	0.215† / -	0.389 / 0.209†	0.403 / 0.294	0.415 / 0.332	0.421 / 0.293	0.392 / 0.357
X1	0.119 / 0.257	0.154 / -	0.269 / 0.221†	0.231 / 0.237	0.286 / 0.281	0.279 / 0.206†	0.276 / 0.280
C_L1b_S1	0.141 / 0.218	0.153 / -	0.279 / 0.163†	0.345 / 0.242	0.360 / 0.270	0.325 / 0.222	0.302 / 0.280
C_L1r_S1	0.131 / 0.301	0.179 / -	0.294 / 0.209†	0.350 / 0.255	0.356 / 0.270	0.346 / 0.227†	0.301 / 0.316
C_S1_L1r	0.135 / 0.282	0.168 / -	0.285 / 0.205	0.326 / 0.267	0.349 / 0.285	0.318 / 0.242	0.301 / 0.291
C_L2_S4	0.176 / 0.291	0.239 / -	0.355 / 0.231†	0.385 / 0.260	0.392 / 0.283	0.383 / 0.309	0.363 / 0.349
C_L1r_S3	0.172 / 0.296	0.201 / -	0.360 / 0.267	0.380 / 0.278	0.387 / 0.315	0.391 / 0.303	0.368 / 0.350
C_S4_S3	0.186 / 0.315	0.223† / -	0.392 / 0.201†	0.403 / 0.282	0.422 / 0.333	0.415 / 0.287	0.398 / 0.352
C3_L1r_S3_S4	0.150 / 0.303	0.181 / -	0.340 / 0.191†	0.377 / 0.268	0.380 / 0.302	0.375 / 0.249†	0.349 / 0.336
C3_S1_S3_S4	0.128 / 0.292	0.160 / -	0.354 / 0.166†	0.369 / 0.234	0.385 / 0.313	0.377 / 0.193†	0.340 / 0.321
C3_L1r_S1_S4	0.126 / 0.299	0.167 / -	0.289 / 0.163†	0.353 / 0.229	0.354 / 0.278	0.325 / 0.181†	0.296 / 0.314
C4_L1r_S1_S3_S4	0.089 / 0.282	0.122 / -	0.278 / 0.183†	0.336 / 0.217	0.356 / 0.288	0.334 / 0.176†	0.290 / 0.296
C_L1r_X1	0.102 / 0.253	0.121 / -	0.184 / 0.210	0.199 / 0.200	0.222 / 0.250	0.189 / 0.176†	0.193 / 0.255
C_X1_S1	0.104 / 0.251	0.131 / -	0.274 / 0.139†	0.230 / 0.201	0.267 / 0.276	0.291 / 0.144†	0.290 / 0.285
C_S2_X1	0.145 / 0.311	0.184 / -	0.313 / 0.153†	0.285 / 0.217	0.332 / 0.305	0.326 / 0.209†	0.317 / 0.326
C_L1r_X1m	0.112 / 0.250	0.116 / -	0.220 / 0.220	0.232 / 0.223	0.249 / 0.250	0.227 / 0.192†	0.215 / 0.253
C_S1_X1s	0.151 / 0.277	0.206 / -	0.334 / 0.144†	0.302 / 0.195	0.355 / 0.288	0.350 / 0.168†	0.338 / 0.286
C3_L1r_S1_X1	0.104 / 0.263	0.127 / -	0.219 / 0.139†	0.222 / 0.154†	0.250 / 0.231	0.242 / 0.097†	0.232 / 0.288
single obfuscation	0.194 / 0.291	0.252 ⁴ / -	0.377 / 0.301 ³	0.389 / 0.277	0.402 / 0.295	0.393 / 0.297	0.380 / 0.331
held-out family	0.119 / 0.257	0.154 / -	0.269 / -	0.231 / 0.237	0.286 / 0.281	0.279 / -	0.276 / 0.280
depth 2, all seen	0.157 / 0.284	0.188 ⁵ / -	0.327 / 0.236 ²	0.365 / 0.264	0.378 / 0.293	0.363 / 0.273 ⁵	0.339 / 0.323
depth 3, all seen	0.135 / 0.298	0.169 / -	0.328 / -	0.366 / 0.244	0.373 / 0.298	0.359 / -	0.328 / 0.324
depth 4, all seen	0.089 / 0.282	0.122 / -	0.278 / -	0.336 / 0.217	0.356 / 0.288	0.334 / -	0.290 / 0.296
depth 2, unseen inside	0.123 / 0.268	0.152 / -	0.265 / 0.215 ²	0.249 / 0.207	0.285 / 0.274	0.277 / -	0.271 / 0.281
depth 3, unseen inside	0.104 / 0.263	0.127 / -	0.219 / -	0.222 / -	0.250 / 0.231	0.242 / -	0.232 / 0.288

Table 6. **Every method against every obfuscation, CodeLlama-13B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.252 / 0.285	0.278 [†] / -	0.469 / 0.269	0.446 / 0.066 [†]	0.462 / 0.231	0.477 / 0.232 [†]	0.477 / 0.339
L1b	0.222 / 0.277	0.220 [†] / -	0.397 / 0.285	0.423 / 0.066 [†]	0.438 / 0.253	0.426 / 0.304	0.400 / 0.318
L1r	0.225 / 0.327	0.233 [†] / -	0.415 / 0.347	0.420 / 0.051 [†]	0.439 / 0.229	0.428 / 0.274	0.425 / 0.366
L2	0.228 / 0.279	0.244 [†] / -	0.407 / 0.232 [†]	0.417 / 0.026 [†]	0.432 / 0.206	0.424 / 0.152 [†]	0.421 / 0.318
S1	0.213 / 0.304	0.216 / -	0.408 / 0.341	0.413 / 0.059 [†]	0.435 / 0.199	0.435 / -	0.435 / 0.363
S2	0.184 / 0.293	0.187 [†] / -	0.422 / 0.317	0.445 / 0.038 [†]	0.469 / 0.238	0.453 / -	0.441 / 0.365
X1	0.149 / 0.258	0.141 [†] / -	0.294 / 0.274	0.254 / 0.016 [†]	0.292 / 0.219	0.313 / -	0.306 / 0.298
C_L1b_S1	0.183 / 0.253	0.159 / -	0.318 / 0.326	0.367 / 0.059 [†]	0.375 / 0.201	0.370 / -	0.329 / 0.341
C_L1r_S1	0.175 / 0.325	0.175 / -	0.318 / 0.360	0.371 / 0.070 [†]	0.376 / 0.203	0.371 / -	0.356 / 0.368
C_S1_L1r	0.172 / 0.323	0.172 / -	0.315 / 0.351	0.359 / 0.086 [†]	0.363 / 0.230	0.370 / -	0.341 / 0.376
C_L2_S4	0.200 / 0.274	0.187 [†] / -	0.389 / 0.186 [†]	0.422 / 0.020 [†]	0.432 / 0.168 [†]	0.421 / -	0.404 / 0.321
C_L1r_S3	0.186 / 0.305	0.194 [†] / -	0.407 / 0.361	0.409 / 0.041 [†]	0.434 / 0.220	0.428 / -	0.417 / 0.366
C_S4_S3	0.192 / 0.293	0.211 [†] / -	0.424 / 0.326	0.436 / 0.026 [†]	0.459 / 0.239	0.464 / -	0.443 / 0.363
C3_L1r_S3_S4	0.166 / 0.310	0.169 [†] / -	0.382 / 0.337	0.410 / 0.032 [†]	0.429 / 0.224	0.416 / -	0.394 / 0.353
C3_S1_S3_S4	0.147 / 0.295	0.158 / -	0.367 / 0.329	0.408 / 0.035 [†]	0.423 / 0.235	0.419 / -	0.387 / 0.347
C3_L1r_S1_S4	0.163 / 0.316	0.156 / -	0.314 / 0.353	0.367 / 0.061 [†]	0.367 / 0.211	0.362 / -	0.346 / 0.370
C4_L1r_S1_S3_S4	0.117 / 0.310	0.117 / -	0.300 / 0.333	0.353 / 0.049 [†]	0.360 / 0.222	0.363 / -	0.320 / 0.353
C_L1r_X1	0.123 / 0.281	0.115 [†] / -	0.199 / 0.254	0.213 / 0.016 [†]	0.236 / 0.170 [†]	0.227 / -	0.217 / 0.266
C_X1_S1	0.138 / 0.289	0.135 / -	0.295 / 0.264	0.270 / 0.007 [†]	0.307 / 0.201 [†]	- / -	0.310 / 0.319
C_S2_X1	0.160 / 0.271	0.167 [†] / -	0.335 / 0.279	0.334 / 0.008 [†]	0.356 / 0.204 [†]	- / -	0.353 / 0.329
C_L1r_X1m	0.138 / 0.291	0.127 [†] / -	0.246 / 0.259	0.248 / 0.020 [†]	0.279 / 0.179 [†]	- / -	0.251 / 0.283
C_S1_X1s	0.179 / 0.301	0.166 / -	0.364 / 0.290	0.345 / 0.025 [†]	0.360 / 0.199 [†]	- / -	0.376 / 0.317
C3_L1r_S1_X1	0.128 / 0.304	0.107 [†] / -	0.233 / 0.293	0.234 / 0.026 [†]	0.249 / 0.185 [†]	- / -	0.251 / 0.313
single obfuscation	0.215 / 0.296	0.216 ¹ / -	0.410 / 0.323 ⁴	0.423 / -	0.442 / 0.225	0.433 / 0.289 ²	0.424 / 0.346
held-out family	0.149 / 0.258	- / -	0.294 / 0.274	0.254 / -	0.292 / 0.219	0.313 / -	0.306 / 0.298
depth 2, all seen	0.185 / 0.295	0.168 ³ / -	0.362 / 0.345 ⁵	0.394 / -	0.407 / 0.219 ⁵	0.404 / -	0.381 / 0.356
depth 3, all seen	0.159 / 0.307	0.157 ² / -	0.354 / 0.340	0.395 / -	0.406 / 0.223	0.399 / -	0.375 / 0.357
depth 4, all seen	0.117 / 0.310	0.117 / -	0.300 / 0.333	0.353 / -	0.360 / 0.222	0.363 / -	0.320 / 0.353
depth 2, unseen inside	0.148 / 0.286	0.151 ² / -	0.288 / 0.269	0.282 / -	0.307 / -	0.227 ¹ / -	0.301 / 0.303
depth 3, unseen inside	0.128 / 0.304	- / -	0.233 / 0.293	0.234 / -	0.249 / -	- / -	0.251 / 0.313

Table 7. **Every method against every obfuscation, CodeLlama-34B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.254 [†] / 0.363	0.178 [†] / -	0.520 / 0.260	0.494 / 0.200 [†]	0.525 / 0.284	0.526 / 0.239 [†]	0.527 / 0.382
L1b	0.205 / 0.327	0.077 [†] / -	0.417 / 0.253	0.469 / 0.174 [†]	0.494 / 0.287	0.472 / 0.259 [†]	0.429 / 0.360
L1r	0.220 / 0.339	0.117 [†] / -	0.469 / 0.255	0.457 / 0.135 [†]	0.487 / 0.256	0.481 / 0.170 [†]	0.467 / 0.373
L2	0.231 / 0.357	0.109 [†] / -	0.466 / 0.280	0.465 / 0.179 [†]	0.478 / 0.276	0.475 / 0.200 [†]	0.463 / 0.378
S1	0.222 / 0.323	0.079 [†] / -	0.472 / 0.305	0.467 / 0.178 [†]	0.496 / 0.264	0.484 / 0.241 [†]	0.456 / 0.390
S2	0.191 / 0.350	0.040 [†] / -	0.484 / 0.354	0.484 / 0.219 [†]	0.525 / 0.306	0.509 / 0.293 [†]	0.484 / 0.437
X1	0.135 / 0.303	0.050 [†] / -	0.325 / 0.253	0.298 / 0.118 [†]	0.334 / 0.249	0.344 / 0.114 [†]	0.328 / 0.338
C_L1b_S1	0.156 / 0.257	0.033 [†] / -	0.322 / 0.272	0.399 / 0.144 [†]	0.421 / 0.258	0.390 / 0.221 [†]	0.331 / 0.358
C_L1r_S1	0.184 / 0.304	0.063 [†] / -	0.351 / 0.285	0.393 / 0.132 [†]	0.427 / 0.200	0.415 / 0.145 [†]	0.362 / 0.375
C_S1_L1r	0.180 / 0.292	0.053 [†] / -	0.367 / 0.307	0.405 / 0.142 [†]	0.425 / 0.216	0.387 / 0.163 [†]	0.377 / 0.371
C_L2_S4	0.191 / 0.344	0.045 [†] / -	0.431 / 0.291	0.454 / 0.170 [†]	0.478 / 0.273	0.452 / 0.243 [†]	0.436 / 0.384
C_L1r_S3	0.169 / 0.333	0.041 [†] / -	0.451 / 0.327	0.454 / 0.167 [†]	0.482 / 0.256	0.471 / 0.197 [†]	0.453 / 0.419
C_S4_S3	0.199 / 0.352	0.041 [†] / -	0.482 / 0.345	0.487 / 0.218 [†]	0.521 / 0.301	0.505 / 0.276 [†]	0.488 / 0.442
C3_L1r_S3_S4	0.148 / 0.325	0.017 [†] / -	0.416 / 0.330	0.452 / 0.182 [†]	0.473 / 0.251	- / 0.201 [†]	0.425 / 0.427
C3_S1_S3_S4	0.128 / 0.330	0.042 [†] / -	0.428 / 0.323	0.454 / 0.208 [†]	0.472 / 0.288	- / 0.235 [†]	0.442 / 0.413
C3_L1r_S1_S4	0.147 / 0.291	0.041 [†] / -	0.338 / 0.311	0.390 / 0.121 [†]	0.412 / 0.199	- / 0.134 [†]	0.354 / 0.381
C4_L1r_S1_S3_S4	0.105 / 0.283	0.033 [†] / -	0.316 / 0.333	0.372 / 0.164 [†]	0.413 / 0.214	- / 0.175 [†]	0.351 / 0.386
C_L1r_X1	0.115 / 0.275	0.044 [†] / -	0.226 / 0.240	0.249 / 0.060 [†]	0.278 / 0.162 [†]	- / -	0.236 / 0.299
C_X1_S1	0.125 / 0.298	0.064 [†] / -	0.317 / 0.275	0.290 / 0.089 [†]	0.339 / 0.216 [†]	- / -	0.317 / 0.345
C_S2_X1	0.176 / 0.378	0.099 [†] / -	0.372 / 0.366	0.355 / 0.196 [†]	0.403 / 0.288	- / -	0.382 / 0.414
C_L1r_X1m	0.128 / 0.269	0.052 [†] / -	0.261 / 0.230	0.275 / 0.077 [†]	0.294 / 0.158 [†]	- / -	0.279 / 0.308
C_S1_X1s	0.176 / 0.346	0.077 [†] / -	0.390 / 0.302	0.382 / 0.141 [†]	0.415 / 0.268	- / -	0.387 / 0.418
C3_L1r_S1_X1	0.118 / 0.304	0.048 [†] / -	0.256 / 0.296	0.262 / 0.084 [†]	0.297 / 0.188 [†]	- / -	0.267 / 0.357
single obfuscation	0.214 / 0.339	- / -	0.462 / 0.289	0.469 / -	0.496 / 0.278	0.484 / -	0.460 / 0.388
held-out family	0.135 / 0.303	- / -	0.325 / 0.253	0.298 / -	0.334 / 0.249	0.344 / -	0.328 / 0.338
depth 2, all seen	0.180 / 0.314	- / -	0.401 / 0.305	0.432 / -	0.459 / 0.251	0.437 / -	0.408 / 0.391
depth 3, all seen	0.141 / 0.315	- / -	0.394 / 0.322	0.432 / -	0.452 / 0.246	- / -	0.407 / 0.407
depth 4, all seen	0.105 / 0.283	- / -	0.316 / 0.333	0.372 / -	0.413 / 0.214	- / -	0.351 / 0.386
depth 2, unseen inside	0.144 / 0.313	- / -	0.313 / 0.283	0.310 / -	0.346 / 0.278 ²	- / -	0.320 / 0.357
depth 3, unseen inside	0.118 / 0.304	- / -	0.256 / 0.296	0.262 / -	0.297 / -	- / -	0.267 / 0.357

Table 8. **Every method against every obfuscation, Llama-3.1-8B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base	ICL	clean LoRA	breadth	anchored	mixture	merge
	out / in	out / in	out / in	out / in	out / in	out / in	out / in
L0	0.258 / 0.409	0.320 / -	0.448 / 0.335	0.426 / 0.320	0.445 / 0.332	0.475 / 0.338	0.462 / 0.414
L1b	0.222 / 0.371	0.276 / -	0.361 / 0.297	0.387 / 0.280	0.407 / 0.285	0.422 / 0.318	0.386 / 0.383
L1r	0.242 / 0.373	0.284 / -	0.387 / 0.312	0.387 / 0.290	0.405 / 0.297	0.407 / 0.311	0.398 / 0.372
L2	0.215 / 0.376	0.278 / -	0.391 / 0.297	0.386 / 0.284	0.409 / 0.294	0.425 / 0.311	0.396 / 0.364
S1	0.213 / 0.375	0.256 / -	0.395 / 0.325	0.391 / 0.285	0.413 / 0.291	0.428 / 0.316	0.412 / 0.411
S2	0.205 / 0.378	0.272 / -	0.413 / 0.345	0.413 / 0.326	0.445 / 0.325	0.456 / 0.366	0.437 / 0.435
X1	0.132 / 0.263	0.161 / -	0.249 / 0.267	0.222 / 0.211	0.255 / 0.228	0.280 / 0.258	0.266 / 0.311
C_L1b_S1	0.152 / 0.287	0.185 / -	0.279 / 0.253	0.338 / 0.274	0.368 / 0.258	0.369 / 0.306	0.307 / 0.352
C_L1r_S1	0.162 / 0.297	0.220 / -	0.303 / 0.281	0.352 / 0.280	0.363 / 0.283	0.366 / 0.293	0.328 / 0.350
C_S1_L1r	0.159 / 0.303	0.208 / -	0.277 / 0.269	0.338 / 0.274	0.354 / 0.274	0.347 / 0.293	0.306 / 0.356
C_L2_S4	0.205 / 0.320	0.257 / -	0.349 / 0.275	0.386 / 0.269	0.401 / 0.286	0.413 / 0.313	0.383 / 0.388
C_L1r_S3	0.198 / 0.342	0.266 / -	0.367 / 0.297	0.382 / 0.299	0.398 / 0.303	0.407 / 0.320	0.387 / 0.375
C_S4_S3	0.214 / 0.367	0.269 / -	0.421 / 0.333	0.411 / 0.313	0.444 / 0.329	0.459 / 0.363	0.437 / 0.417
C3_L1r_S3_S4	0.183 / 0.309	0.233 / -	0.329 / 0.271	0.374 / 0.303	0.391 / 0.300	0.399 / 0.317	0.370 / 0.372
C3_S1_S3_S4	0.160 / 0.353	0.220 / -	0.359 / 0.311	0.376 / 0.289	0.393 / 0.281	0.415 / 0.311	0.377 / 0.408
C3_L1r_S1_S4	0.160 / 0.302	0.208 / -	0.279 / 0.270	0.342 / 0.262	0.358 / 0.271	0.353 / 0.276	0.319 / 0.341
C4_L1r_S1_S3_S4	0.140 / 0.281	0.191 / -	0.263 / 0.234	0.335 / 0.280	0.358 / 0.279	0.342 / 0.274	0.310 / 0.340
C_L1r_X1	0.100 / 0.218	0.118 / -	0.165 / 0.220	0.181 / 0.193	0.207 / 0.239	0.204 / 0.230	0.184 / 0.275
C_X1_S1	0.130 / 0.291	0.179 / -	0.271 / 0.283	0.225 / 0.210	0.253 / 0.220	0.294 / 0.269	0.296 / 0.332
C_S2_X1	0.170 / 0.323	0.212 / -	0.296 / 0.288	0.284 / 0.229	0.317 / 0.241	0.336 / 0.284	0.317 / 0.350
C_L1r_X1m	0.115 / 0.232	0.148 / -	0.184 / 0.210	0.216 / 0.217	0.232 / 0.245	0.238 / 0.225	0.217 / 0.274
C_S1_X1s	0.193 / 0.357	0.231 / -	0.341 / 0.287	0.299 / 0.205	0.335 / 0.212	0.362 / 0.276	0.349 / 0.376
C3_L1r_S1_X1	0.105 / 0.268	0.156 / -	0.194 / 0.234	0.207 / 0.188	0.224 / 0.194	0.238 / 0.241	0.215 / 0.305
single obfuscation	0.219 / 0.375	0.273 / -	0.390 / 0.315	0.393 / 0.293	0.416 / 0.299	0.427 / 0.324	0.406 / 0.393
held-out family	0.132 / 0.263	0.161 / -	0.249 / 0.267	0.222 / 0.211	0.255 / 0.228	0.280 / 0.258	0.266 / 0.311
depth 2, all seen	0.182 / 0.319	0.234 / -	0.332 / 0.284	0.368 / 0.285	0.388 / 0.289	0.394 / 0.315	0.358 / 0.373
depth 3, all seen	0.168 / 0.321	0.221 / -	0.322 / 0.284	0.364 / 0.285	0.380 / 0.284	0.389 / 0.301	0.355 / 0.374
depth 4, all seen	0.140 / 0.281	0.191 / -	0.263 / 0.234	0.335 / 0.280	0.358 / 0.279	0.342 / 0.274	0.310 / 0.340
depth 2, unseen inside	0.142 / 0.284	0.177 / -	0.252 / 0.258	0.241 / 0.211	0.269 / 0.231	0.287 / 0.257	0.273 / 0.321
depth 3, unseen inside	0.105 / 0.268	0.156 / -	0.194 / 0.234	0.207 / 0.188	0.224 / 0.194	0.238 / 0.241	0.215 / 0.305

Table 9. **Every method against every obfuscation, StarCoder2-15B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.000 [†] / 0.366	0.008 [†] / -	0.542 / 0.411	0.529 / 0.311	0.532 / 0.399	0.546 / 0.435	0.539 / 0.452
L1b	0.000 [†] / 0.216 [†]	0.000 [†] / -	0.431 / 0.316	0.487 / 0.301	0.510 / 0.395	0.489 / 0.382	0.426 / 0.364
L1r	0.000 [†] / 0.282 [†]	0.000 [†] / -	0.483 / 0.354	0.485 / 0.273	0.509 / 0.369	0.492 / 0.368	0.476 / 0.423
L2	0.000 [†] / 0.355	0.002 [†] / -	0.479 / 0.391	0.486 / 0.297	0.503 / 0.379	0.493 / 0.406	0.466 / 0.421
S1	0.000 [†] / 0.235 [†]	0.010 [†] / -	0.508 / 0.361	0.487 / 0.257 [†]	0.501 / 0.364	0.510 / 0.377	0.487 / 0.421
S2	0.000 [†] / 0.238 [†]	0.023 [†] / -	0.512 / 0.359	0.514 / 0.300	0.539 / 0.384	0.533 / 0.404	0.518 / 0.440
X1	0.000 [†] / 0.235 [†]	0.009 [†] / -	0.347 / 0.254 [†]	0.320 / 0.177 [†]	0.343 / 0.281	0.353 / 0.261 [†]	0.343 / 0.328
C_L1b_S1	0.000 [†] / 0.122 [†]	0.001 [†] / -	0.360 / 0.254 [†]	0.416 / 0.211 [†]	0.456 / 0.317	0.417 / 0.297	0.354 / 0.324
C_L1r_S1	0.000 [†] / 0.165 [†]	0.000 [†] / -	0.393 / 0.301	0.423 / 0.214 [†]	0.440 / 0.303	0.419 / 0.305	0.389 / 0.388
C_S1_L1r	0.000 [†] / 0.136 [†]	0.000 [†] / -	0.377 / 0.311	0.406 / 0.183 [†]	0.431 / 0.303	0.418 / 0.309	0.387 / 0.374
C_L2_S4	0.000 [†] / 0.215 [†]	0.005 [†] / -	0.454 / 0.342	0.470 / 0.260 [†]	0.508 / 0.353	0.485 / 0.378	0.438 / 0.395
C_L1r_S3	0.000 [†] / 0.243 [†]	0.000 [†] / -	0.473 / 0.300	0.483 / 0.282	0.504 / 0.368	0.490 / 0.323	0.459 / 0.383
C_S4_S3	0.000 [†] / 0.243 [†]	0.028 [†] / -	0.511 / 0.368	0.516 / 0.301	0.539 / 0.386	0.525 / 0.389	0.506 / 0.448
C3_L1r_S3_S4	0.000 [†] / 0.182 [†]	0.004 [†] / -	0.448 / 0.294	0.484 / 0.242 [†]	0.503 / 0.339	0.475 / 0.315	0.436 / 0.386
C3_S1_S3_S4	0.000 [†] / 0.191 [†]	0.019 [†] / -	0.476 / 0.337	0.465 / 0.235 [†]	0.493 / 0.333	0.480 / 0.345	0.464 / 0.406
C3_L1r_S1_S4	0.000 [†] / 0.096 [†]	0.002 [†] / -	0.370 / 0.293	0.417 / 0.209 [†]	0.439 / 0.270	0.413 / 0.297	0.376 / 0.353
C4_L1r_S1_S3_S4	0.000 [†] / 0.104 [†]	0.001 [†] / -	0.357 / 0.289	0.412 / 0.206 [†]	0.437 / 0.273	0.401 / 0.271	0.358 / 0.358
C_L1r_X1	0.000 [†] / 0.123 [†]	0.002 [†] / -	0.248 / 0.165 [†]	0.274 / 0.165 [†]	0.286 / 0.205 [†]	0.278 / 0.168 [†]	0.259 / 0.262
C_X1_S1	0.000 [†] / 0.194 [†]	0.022 [†] / -	0.357 / 0.259 [†]	0.322 / 0.127 [‡]	0.342 / 0.243 [†]	0.385 / 0.238 [†]	0.371 / 0.355
C_S2_X1	0.000 [†] / 0.184 [†]	0.014 [†] / -	0.403 / 0.261 [†]	0.375 / 0.196 [†]	0.397 / 0.268 [†]	0.413 / 0.276 [†]	0.403 / 0.384
C_L1r_X1m	0.000 [†] / 0.179 [†]	0.003 [†] / -	0.307 / 0.191 [†]	0.323 / 0.178 [†]	0.338 / 0.234	0.328 / 0.180 [†]	0.317 / 0.267
C_S1_X1s	0.000 [†] / 0.258 [†]	0.044 [†] / -	0.434 / 0.291 [†]	0.408 / 0.176 [†]	0.434 / 0.286 [†]	0.443 / 0.272 [†]	0.427 / 0.384
C3_L1r_S1_X1	0.000 [†] / 0.153 [†]	0.002 [†] / -	0.278 / 0.191 [†]	0.298 / 0.107 [‡]	0.302 / 0.167 [†]	0.306 / 0.146 [†]	0.286 / 0.323
single obfuscation	- / 0.355 ¹	- / -	0.482 / 0.356	0.492 / 0.293 ⁴	0.512 / 0.378	0.503 / 0.387	0.475 / 0.414
held-out family	- / -	- / -	0.347 / -	0.320 / -	0.343 / 0.281	0.353 / -	0.343 / 0.328
depth 2, all seen	- / -	- / -	0.428 / 0.325 ⁵	0.452 / 0.291 ²	0.480 / 0.338	0.459 / 0.334	0.422 / 0.385
depth 3, all seen	- / -	- / -	0.431 / 0.308	0.455 / -	0.478 / 0.314	0.456 / 0.319	0.425 / 0.382
depth 4, all seen	- / -	- / -	0.357 / 0.289	0.412 / -	0.437 / 0.273	0.401 / 0.271	0.358 / 0.358
depth 2, unseen inside	- / -	- / -	0.350 / -	0.341 / -	0.359 / 0.234 ¹	0.369 / -	0.355 / 0.330
depth 3, unseen inside	- / -	- / -	0.278 / -	0.298 / -	0.302 / -	0.306 / -	0.286 / 0.323

Table 10. **Every method against every obfuscation, Gemma-3-12B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.334 [†] / 0.453	0.340 [†] / -	0.520 / 0.440	0.507 / 0.447	0.502 / 0.355	0.530 / 0.474	0.523 / 0.481
L1b	0.262 [†] / 0.405	0.274 [†] / -	0.441 / 0.353	0.475 / 0.427	0.488 / 0.309 [†]	0.481 / 0.436	0.464 / 0.425
L1r	0.273 [†] / 0.420	0.283 [†] / -	0.459 / 0.394	0.463 / 0.411	0.472 / 0.280 [†]	0.477 / 0.433	0.474 / 0.440
L2	0.269 [†] / 0.403	0.281 [†] / -	0.456 / 0.378	0.471 / 0.405	0.478 / 0.295 [†]	0.481 / 0.424	0.465 / 0.420
S1	0.269 [†] / 0.424	0.289 [†] / -	0.456 / 0.375	0.457 / 0.421	0.483 / 0.256 [†]	0.496 / 0.446	0.479 / 0.435
S2	0.290 [†] / 0.443	0.308 [†] / -	0.500 / 0.444	0.503 / 0.457	0.504 / 0.277 [†]	0.527 / 0.486	0.513 / 0.484
X1	0.152 / 0.333	0.183 / -	0.310 / 0.305	0.263 / 0.333	0.289 / 0.199 [†]	0.314 / -	0.311 / 0.371
C_L1b_S1	0.195 [†] / 0.349	0.197 [†] / -	0.333 / 0.317	0.406 / 0.366	0.420 / 0.214 [†]	0.414 / -	0.359 / 0.396
C_L1r_S1	0.200 [†] / 0.356	0.204 [†] / -	0.336 / 0.319	0.389 / 0.379	0.416 / 0.203 [†]	0.398 / -	0.362 / 0.386
C_S1_L1r	0.181 [†] / 0.337	0.193 [†] / -	0.334 / 0.319	0.393 / 0.379	0.414 / 0.207 [†]	0.389 / -	0.338 / 0.405
C_L2_S4	0.249 [†] / 0.387	0.262 [†] / -	0.438 / 0.381	0.460 / 0.416	0.469 / 0.222 [†]	0.459 / -	0.452 / 0.430
C_L1r_S3	0.260 [†] / 0.427	0.274 [†] / -	0.438 / 0.402	0.462 / 0.423	0.469 / 0.254 [†]	- / -	0.450 / 0.443
C_S4_S3	0.297 [†] / 0.434	0.308 [†] / -	0.491 / 0.438	0.506 / 0.462	0.503 / 0.274 [†]	- / -	0.496 / 0.472
C3_L1r_S3_S4	0.232 [†] / 0.398	0.256 [†] / -	0.419 / 0.386	0.454 / 0.420	0.467 / 0.223 [†]	- / -	0.440 / 0.441
C3_S1_S3_S4	0.220 [†] / 0.387	0.256 [†] / -	0.425 / 0.354	0.452 / 0.419	0.465 / 0.240 [†]	- / -	0.449 / 0.430
C3_L1r_S1_S4	0.176 [†] / 0.347	0.191 [†] / -	0.320 / 0.302	0.390 / 0.376	0.415 / 0.181 [†]	- / -	0.341 / 0.377
C4_L1r_S1_S3_S4	0.170 [†] / 0.329	0.172 [†] / -	0.302 / 0.293	0.394 / 0.364	0.413 / 0.180 [†]	- / -	0.332 / 0.370
C_L1r_X1	0.102 / 0.258	0.126 / -	0.195 / 0.242	0.206 / 0.279	0.224 / 0.121 [†]	- / -	0.203 / 0.274
C_X1_S1	0.172 / 0.348	0.193 / -	0.317 / 0.282	0.271 / 0.331	0.296 / 0.132 [†]	- / -	0.312 / 0.375
C_S2_X1	0.201 / 0.360	0.222 / -	0.368 / 0.339	0.349 / 0.357	0.364 / 0.210 [†]	- / -	0.368 / 0.395
C_L1r_X1m	0.108 [†] / 0.273	0.115 [†] / -	0.220 / 0.269	0.235 / 0.323	0.244 / 0.146 [†]	- / -	0.231 / 0.285
C_S1_X1s	0.255 / 0.412	0.275 / -	0.395 / 0.371	0.374 / 0.383	0.395 / 0.178 [†]	- / -	0.393 / 0.438
C3_L1r_S1_X1	0.136 / 0.279	0.147 [†] / -	0.225 / 0.226	0.236 / 0.270	0.258 / 0.115 [†]	- / -	0.246 / 0.300
single obfuscation	- / 0.419	- / -	0.462 / 0.389	0.474 / 0.424	0.485 / -	0.492 / 0.445	0.479 / 0.441
held-out family	0.152 / 0.333	0.183 / -	0.310 / 0.305	0.263 / 0.333	0.289 / -	0.314 / -	0.311 / 0.371
depth 2, all seen	- / 0.382	- / -	0.395 / 0.363	0.436 / 0.404	0.448 / -	0.415 [‡] / -	0.409 / 0.422
depth 3, all seen	- / 0.377	- / -	0.388 / 0.348	0.432 / 0.405	0.449 / -	- / -	0.410 / 0.416
depth 4, all seen	- / 0.329	- / -	0.302 / 0.293	0.394 / 0.364	0.413 / -	- / -	0.332 / 0.370
depth 2, unseen inside	0.182 [‡] / 0.330	0.204 [‡] / -	0.299 / 0.301	0.287 / 0.335	0.305 / -	- / -	0.301 / 0.353
depth 3, unseen inside	0.136 / 0.279	- / -	0.225 / 0.226	0.236 / 0.270	0.258 / -	- / -	0.246 / 0.300

Table 11. **Every method against every obfuscation, CodeGemma-7B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.237 / 0.266	0.267 [†] / -	0.465 / 0.364	0.439 / 0.324	0.445 / 0.098 [‡]	0.474 / 0.323	0.465 / 0.329
L1b	0.185 / 0.268	0.209 [†] / -	0.382 / 0.309	0.425 / 0.298	0.435 / 0.067 [‡]	0.443 / 0.307	0.391 / 0.301
L1r	0.213 / 0.251	0.225 [†] / -	0.399 / 0.333	0.417 / 0.301	0.417 / 0.107 [‡]	0.426 / 0.294	0.411 / 0.303
L2	0.208 / 0.239	0.230 [†] / -	0.412 / 0.327	0.422 / 0.291	0.422 / 0.078 [‡]	0.440 / 0.287	0.406 / 0.299
S1	0.195 / 0.234	0.212 / -	0.405 / 0.357	0.415 / 0.337	0.421 / 0.125 [‡]	0.457 / 0.320	0.414 / 0.312
S2	0.198 / 0.270	0.242 [†] / -	0.450 / 0.372	0.429 / 0.351	0.448 / 0.126 [‡]	0.479 / 0.359	0.448 / 0.330
X1	0.143 / 0.190	0.157 / -	0.271 / 0.260	0.251 / 0.217	0.247 / 0.049 [‡]	- / -	0.274 / 0.264
C_L1b_S1	0.148 / 0.229	0.163 / -	0.286 / 0.274	0.372 / 0.276	0.368 / 0.089 [‡]	- / -	0.302 / 0.297
C_L1r_S1	0.144 / 0.170	0.144 / -	0.315 / 0.289	0.373 / 0.290	0.367 / 0.086 [‡]	- / -	0.318 / 0.271
C_S1_L1r	0.132 / 0.168	0.149 / -	0.288 / 0.284	0.363 / 0.316	0.361 / 0.099 [‡]	- / -	0.300 / 0.281
C_L2_S4	0.176 / 0.239	0.208 [†] / -	0.391 / 0.318	0.423 / 0.299	0.421 / 0.070 [‡]	- / -	0.391 / 0.295
C_L1r_S3	0.183 / 0.223	0.216 [†] / -	0.396 / 0.326	0.415 / 0.314	0.410 / 0.150 [†]	- / -	0.393 / 0.293
C_S4_S3	0.202 / 0.257	0.235 [†] / -	0.448 / 0.372	0.430 / 0.357	0.441 / 0.130 [‡]	- / -	0.445 / 0.348
C3_L1r_S3_S4	0.157 / 0.187	0.182 [†] / -	0.376 / 0.318	0.407 / 0.312	0.411 / 0.083 [‡]	- / -	0.372 / 0.294
C3_S1_S3_S4	0.158 / 0.231	0.185 / -	0.367 / 0.338	0.395 / 0.332	0.405 / 0.138 [‡]	- / -	0.378 / 0.306
C3_L1r_S1_S4	0.126 / 0.173	0.137 / -	0.289 / 0.274	0.367 / 0.286	0.366 / 0.064 [‡]	- / -	0.303 / 0.267
C4_L1r_S1_S3_S4	0.124 / 0.150	0.125 [†] / -	0.271 / 0.267	0.361 / 0.290	0.359 / 0.081 [‡]	- / -	0.303 / 0.270
C_L1r_X1	0.118 / 0.133 [†]	0.110 / -	0.194 / 0.208	0.212 / 0.174	0.197 / 0.036 [‡]	- / -	0.184 / 0.202
C_X1_S1	0.144 / 0.202	0.152 / -	0.285 / 0.303	0.261 / 0.209	0.247 / 0.034 [‡]	- / -	0.288 / 0.280
C_S2_X1	0.188 / 0.227	0.198 / -	0.344 / 0.300	0.323 / 0.241	0.326 / 0.031 [‡]	- / -	0.340 / 0.293
C_L1r_X1m	0.130 / 0.146 [†]	0.112 / -	0.214 / 0.231	0.256 / 0.211	0.234 / 0.049 [‡]	- / -	0.201 / 0.230
C_S1_X1s	0.186 / 0.230	0.210 / -	0.338 / 0.322	0.351 / 0.230	0.339 / 0.037 [‡]	- / -	0.358 / 0.302
C3_L1r_S1_X1	0.134 / 0.168	0.118 / -	0.218 / 0.230	0.241 / 0.140 [†]	0.246 / 0.013 [‡]	- / -	0.224 / 0.232
single obfuscation	0.200 / 0.252	0.212 ¹ / -	0.410 / 0.340	0.422 / 0.315	0.429 / -	0.449 / 0.313	0.414 / 0.309
held-out family	0.143 / 0.190	0.157 / -	0.271 / 0.260	0.251 / 0.217	0.247 / -	- / -	0.274 / 0.264
depth 2, all seen	0.164 / 0.214	0.152 ³ / -	0.354 / 0.310	0.396 / 0.308	0.395 / -	- / -	0.358 / 0.298
depth 3, all seen	0.147 / 0.197	0.161 ² / -	0.344 / 0.310	0.390 / 0.310	0.394 / -	- / -	0.351 / 0.289
depth 4, all seen	0.124 / 0.150	- / -	0.271 / 0.267	0.361 / 0.290	0.359 / -	- / -	0.303 / 0.270
depth 2, unseen inside	0.153 / 0.220 ³	0.156 / -	0.275 / 0.273	0.280 / 0.213	0.269 / -	- / -	0.274 / 0.261
depth 3, unseen inside	0.134 / 0.168	0.118 / -	0.218 / 0.230	0.241 / -	0.246 / -	- / -	0.224 / 0.232

Table 12. **Every method against every obfuscation, Granite-3.1-8B.** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy **above** or **below** base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.285 / 0.291	0.293 / -	0.430 / 0.206 [†]	0.416 / 0.261	0.396 / 0.117 [†]	0.438 / 0.268	0.433 / 0.304
L1b	0.267 / 0.255	0.267 / -	0.338 / 0.156 [†]	0.391 / 0.228	0.381 / 0.117 [†]	0.396 / 0.252	0.350 / 0.278
L1r	0.268 / 0.284	0.260 / -	0.375 / 0.209 [†]	0.383 / 0.268	0.378 / 0.150	0.391 / 0.280	0.384 / 0.299
L2	0.251 / 0.247	0.248 / -	0.372 / 0.209 [†]	0.387 / 0.236	0.377 / 0.133 [†]	0.400 / 0.244	0.378 / 0.294
S1	0.254 / 0.305	0.253 / -	0.346 / 0.196 [†]	0.379 / 0.220	0.358 / 0.099 [†]	0.403 / 0.260	0.371 / 0.284
S2	0.272 / 0.329	0.276 / -	0.386 / 0.182 [†]	0.415 / 0.244	0.394 / 0.098 [†]	0.422 / 0.278	0.402 / 0.315
X1	0.147 / 0.198 [†]	0.160 / -	0.216 / 0.123 [†]	0.207 / 0.156 [†]	0.188 / 0.048 [‡]	0.244 / 0.210	0.231 / 0.243
C_L1b_S1	0.219 / 0.216	0.191 / -	0.263 / 0.136 [†]	0.342 / 0.199	0.337 / 0.082 [†]	0.324 / 0.234	0.279 / 0.228
C_L1r_S1	0.204 / 0.284	0.187 / -	0.279 / 0.206	0.344 / 0.225	0.338 / 0.126 [†]	0.334 / 0.250	0.301 / 0.267
C_S1_L1r	0.224 / 0.262	0.187 / -	0.279 / 0.208	0.326 / 0.241	0.319 / 0.132 [†]	0.327 / 0.254	0.293 / 0.276
C_L2_S4	0.242 / 0.278	0.245 / -	0.334 / 0.182 [†]	0.376 / 0.225	0.376 / 0.117 [†]	0.384 / 0.242	0.351 / 0.287
C_L1r_S3	0.259 / 0.296	0.259 / -	0.350 / 0.160 [†]	0.382 / 0.240	0.376 / 0.145 [†]	0.378 / 0.279	0.365 / 0.296
C_S4_S3	0.270 / 0.334	0.282 / -	0.380 / 0.180 [†]	0.412 / 0.254	0.389 / 0.100 [†]	0.429 / 0.290	0.394 / 0.313
C3_L1r_S3_S4	0.257 / 0.331	0.245 / -	0.333 / 0.180 [†]	0.380 / 0.238	0.375 / 0.135 [†]	0.375 / 0.279	0.348 / 0.276
C3_S1_S3_S4	0.242 / 0.323	0.248 / -	0.316 / 0.181 [†]	0.372 / 0.214	0.358 / 0.062 [†]	0.380 / 0.273	0.342 / 0.289
C3_L1r_S1_S4	0.221 / 0.293	0.200 / -	0.279 / 0.192	0.333 / 0.223	0.330 / 0.111 [†]	- / 0.246	0.287 / 0.262
C4_L1r_S1_S3_S4	0.203 / 0.305	0.205 / -	0.258 / 0.175 [†]	0.325 / 0.216	0.334 / 0.106 [†]	- / 0.253	0.281 / 0.272
C_L1r_X1	0.136 / 0.165 [†]	0.124 / -	0.157 / 0.102 [†]	0.172 / 0.163	0.163 / 0.072 [†]	- / 0.170	0.173 / 0.201
C_X1_S1	0.153 / 0.234	0.169 / -	0.222 / 0.156 [†]	0.203 / 0.161	0.186 / 0.034 [‡]	- / 0.228	0.244 / 0.251
C_S2_X1	0.209 / 0.300	0.217 / -	0.263 / 0.151 [†]	0.252 / 0.150 [†]	0.240 / 0.043 [‡]	- / 0.249	0.272 / 0.277
C_L1r_X1m	0.158 / 0.159 [†]	0.145 / -	0.188 / 0.117 [†]	0.204 / 0.180	0.183 / 0.091 [†]	- / 0.199	0.211 / 0.196
C_S1_X1s	0.206 / 0.282	0.215 / -	0.298 / 0.141 [†]	0.267 / 0.173	0.269 / 0.037 [‡]	- / 0.232	0.293 / 0.268
C3_L1r_S1_X1	0.167 / 0.259	0.147 / -	0.186 / 0.163 [†]	0.195 / 0.156	0.198 / 0.057 [‡]	- / 0.219	0.217 / 0.239
single obfuscation	0.262 / 0.284	0.261 / -	0.363 / -	0.391 / 0.239	0.378 / 0.150 ¹	0.402 / 0.263	0.377 / 0.294
held-out family	0.147 / -	0.160 / -	0.216 / -	0.207 / -	0.188 / -	0.244 / 0.210	0.231 / 0.243
depth 2, all seen	0.236 / 0.278	0.225 / -	0.314 / 0.207 ²	0.364 / 0.231	0.356 / -	0.363 / 0.258	0.330 / 0.278
depth 3, all seen	0.240 / 0.316	0.231 / -	0.309 / 0.192 ¹	0.362 / 0.225	0.354 / -	0.377 ² / 0.266	0.326 / 0.276
depth 4, all seen	0.203 / 0.305	0.205 / -	0.258 / -	0.325 / 0.216	0.334 / -	- / 0.253	0.281 / 0.272
depth 2, unseen inside	0.172 / 0.272 ³	0.174 / -	0.226 / -	0.220 / 0.169 ⁴	0.208 / -	- / 0.216	0.239 / 0.239
depth 3, unseen inside	0.167 / 0.259	0.147 / -	0.186 / -	0.195 / 0.156	0.198 / -	- / 0.219	0.217 / 0.239

Table 13. **Every method against every obfuscation, Qwen2.5-Coder-1.5B (pilot).** Each cell is *output prediction / input prediction*: raw exact-match accuracy on the forward task (predict the return value of the code as shown) and, after the slash, on the backward task (produce a call that returns the shown value), on the held-out programs of that condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them. **Colour**: an accuracy *above* or *below* base in the same direction on the same condition; base is the reference, and an uncoloured number means no comparison was available rather than a tie. A **red number carrying † or ‡** is red for the other reason — its format-failure rate is above 0.25, so the accuracy beneath it is not a fair reading of the arm — and those cells are excluded from the bold means. **Bold rows** are means over the group named, equally weighted per condition; a superscript gives the number of readable cells averaged when it is short of the group. The family adapter (tuned_X1) is deliberately not a column: it is trained on the held-out family’s sibling, so on exactly the rows that matter most it is the one arm for which that family is not held out; the **held-out family** row names a group of conditions, not an arm. **Backward is graded by execution**: any call returning the gold value is correct, because inversion is many-to-one. †: format-failure rate above 0.25, mostly replies that do not meet the output contract. ‡: above 0.25 backward, but most replies are exactly the gold *forward* answer — the arm answered the other question, which is a result rather than a broken template. **This model is the pilot**: it predates X1, the depth-3/4 stacks and the unseen-containing stacks, and the in-context, anchored and family arms were never built for it, so those rows and columns are absent rather than empty; it does carry S3/S4 as standalone transforms, which the panel models do not. ‘-’: not run.

Condition	base out / in	ICL out / in	clean LoRA out / in	breadth out / in	anchored out / in	mixture out / in	merge out / in
L0	0.290 / 0.335	- / -	0.444 / 0.253	0.416 / 0.211	- / -	0.494 / -	0.494 / 0.334
L1b	0.233 / 0.274	- / -	0.343 / 0.205	0.381 / 0.214	- / -	0.477 / -	0.398 / 0.257
L1r	0.233 / 0.294	- / -	0.363 / 0.217	0.367 / 0.215	- / -	0.511 / -	0.420 / 0.270
L2	0.210 / 0.311	- / -	0.374 / 0.247	0.379 / 0.214	- / -	0.523 / -	0.500 / 0.300
S1	0.262 / 0.311	- / -	0.389 / 0.268	0.392 / 0.241	- / -	0.476 / -	0.441 / 0.322
S2	0.188 / 0.338	- / -	0.419 / 0.273	0.415 / 0.218	- / -	0.494 / -	0.443 / 0.316
S3	0.182 / 0.330	- / -	0.431 / 0.279	0.416 / 0.254	- / -	0.443 / -	0.466 / 0.318
S4	0.222 / 0.353	- / -	0.432 / 0.254	0.415 / 0.194	- / -	0.489 / -	0.455 / 0.306
C_L1b_S1	0.320 / 0.240	- / -	0.248 / 0.191	0.322 / 0.216	- / -	0.420 / -	0.407 / 0.217
C_L1r_S1	0.267 / 0.281	- / -	0.271 / 0.215	0.334 / 0.218	- / -	0.407 / -	0.400 / 0.246
C_S1_L1r	0.247 / 0.266	- / -	0.246 / 0.203	0.326 / 0.228	- / -	0.420 / -	0.440 / 0.221
C_L2_S4	0.176 / 0.304	- / -	0.344 / 0.215	0.373 / 0.192	- / -	0.540 / -	0.443 / 0.263
C_L1r_S3	0.159 / 0.261	- / -	0.345 / 0.213	0.368 / 0.233	- / -	0.460 / -	0.432 / 0.259
C_S4_S3	0.182 / 0.354	- / -	0.423 / 0.281	0.413 / 0.220	- / -	0.477 / -	0.466 / 0.308
single obfuscation	0.218 / 0.316	- / -	0.393 / 0.249	0.395 / 0.221	- / -	0.488 / -	0.446 / 0.299
depth 2, all seen	0.225 / 0.284	- / -	0.313 / 0.220	0.356 / 0.218	- / -	0.454 / -	0.431 / 0.252

5 RQ1: Do current adaptation methods generalise?

Adapting a code model to obfuscated input works, and works easily: tuning on output prediction over obfuscated programs gains roughly **19 points** over the untuned model on the transforms it was trained on. We apply the two tests of Section 3 to the field’s existing answers. These are baselines, not proposals. Merging and breadth fail both tests; routing tracks breadth wherever the two are measured together and inherits its verdict.

Routing composes exactly as far as breadth does, and merging does not compose at all (Table 31, Appendix D). On the ten stacks of seen transforms, a learned router beats a *random* gate by +1.34 [+0.48, +2.17]: the mixture is worth the whole gain (+19.79 over the untuned model) and the routing about a point of it. A hard argmax router is indistinguishable from the soft one (+0.03 [−0.13, +0.19]), so a stacked input gives the mixture no blending advantage, and the router matches breadth training, −0.28 [−1.57, +1.09]. The best merge (DARE-TIES) matches the clean-code control, +0.52 [−0.15, +1.20], the others fall 1.7 to 20.6 points below it, while breadth stands +3.67 [+2.02, +5.24] above the same control.

Table 32 (Appendix D) reports the same merge arms measured across the panel.

Breadth is the interesting case. It composes on stacks built from transforms it saw, and the advantage evaporates the moment a stack contains a family it did not. Table 14 states this as a *difference*, because one interval being positive while another is negative is not a test of the difference between them: breadth’s advantage over clean-code tuning is **2.7 to 5.3 points larger** on seen stacks than on stacks containing an unseen family, and the gap is significant on **every one of the six models** we could measure it on, across four lineages. What breadth buys is recombination of what the training data covered, bounded by that coverage.

Table 14. **RQ1: the dissociation, measured within model on one program set.** Breadth (mono_all) against the clean-code control (tuned_L0) on stacks built from *seen* transforms and on stacks containing an *unseen* family. The *difference* is the measured quantity; the two levels are context, and on most models the unseen-side interval straddles zero on its own. Both halves are paired on the 319 programs common to every cell of both groups. Bold intervals exclude zero.

Model	Lineage	Seen stacks	Unseen inside	Difference
CodeLlama-7B	Meta	+3.36 [+1.57, +5.27]	−1.92 [−3.70, −0.14]	+5.28 [+3.20, +7.54]
CodeLlama-34B	Meta	+2.27 [+0.44, +4.09]	−0.77 [−2.62, +1.12]	+3.03 [+0.78, +5.28]
Llama-3.1-8B	Meta	+2.30 [+0.47, +4.15]	−1.05 [−2.72, +0.63]	+3.35 [+1.50, +5.30]
StarCoder2-15B	BigCode	+1.50 [−0.02, +3.00]	−1.19 [−2.93, +0.49]	+2.68 [+0.77, +4.65]
Gemma-3-12B	Google	+2.63 [+0.56, +4.64]	−2.09 [−3.81, −0.42]	+4.72 [+2.62, +6.85]
Granite-3.1-8B	IBM	+3.66 [+1.54, +5.68]	−0.24 [−2.02, +1.57]	+3.91 [+1.41, +6.32]
difference significant on				6 of 6 models

The reversal test is independent and gives the same answer. Table 33 (Appendix D) reports the *direction ratio* — backward gain divided by forward gain, both against the untuned model — for the same adapters on the same programs asked backwards. **Every method that fits the forward task directly is negative:** the format-only control at −0.30, clean-code tuning at −0.08, breadth at −0.05; these arms bought forward accuracy by losing backward competence on the very same programs. Both tests point at one explanation: what is acquired is competence at *this task on these surfaces*. Three further instruments, sharing no machinery, say which surface. Breadth’s advantage on a stack depends on the stack still containing an *identifier* transform rather than on its depth or the number of mechanisms in it, and the rule defining that split was fixed before the unseen-containing stimulus existed, which makes its second row an out-of-sample test: the difference is +3.39 on

all-seen stacks and grows to +5.85 on stacks containing the unseen family. Splitting the held-out family into its two mechanisms, the halves **do not transfer to each other** (+1.49 [−0.95, +3.93] and −0.10 [−2.38, +2.19]) yet **either alone recovers about 90 %** of the whole adapter’s gain on the full family — composable skills would deliver half each. And breadth’s log-probability of the gold answer sits about 4 nats below the clean-code control with the entire deficit in the items it gets wrong: it does not become a worse reader, it becomes a more committed guesser. Adaptation anchors on the surface a transform leaves behind, which is exactly what an obfuscator is free to change. The remedy follows: anchor the model to something the transform *cannot* change — its own behaviour on the unobfuscated parent, which exists for every training row by construction. Section 6 is that objective.

Table 15. **RQ2: breadth’s stack gain depends on an identifier transform being present.** Breadth over the clean-code control, split by whether the stack contains an identifier transform, with the *difference* between the groups. The rule defining the split was fixed before the unseen-containing stimulus existed, which makes the second row an out-of-sample test of it. We report the difference because one interval excluding zero while another does not is not a test of the difference between them.

Stimulus	With identifier	Structural only	Difference
All-seen depth-3/4 stacks (in-sample)	+5.00 [+3.02, +6.88]	+1.61 [−1.02, +4.23]	+3.39 [+0.82, +6.16]
Stacks containing the unseen family (out-of-sample)	+1.98 [−0.12, +4.18]	−3.88 [−6.08, −1.71]	+5.85 [+3.10, +8.72]

Takeaway: RQ1

Routing, merging and breadth are baselines, and none survives leaving its training distribution intact. Breadth’s advantage over clean-code tuning is 2.7 to 5.3 points larger on stacks of seen transforms than on stacks containing an unseen family, significant on six of six models across four lineages; and every method that fits the forward task has a negative direction ratio, buying forward accuracy with backward competence on the same programs. Three independent instruments locate what is learned instead: the surface a transform leaves behind — which is exactly what an obfuscator is free to change, and which names the remedy.

6 RQ2: Anchoring to clean-code behaviour

The anchored objective (Section 3, Eq. 1) supervises the student on obfuscated input against a frozen clean-code-tuned teacher reading the unobfuscated parent, on exactly the rows breadth trains on.

The ablation is the contribution. Replacing the clean-code-tuned teacher with an untuned base model makes the arm **collapse** (−8.98 [−11.44, −6.43] against breadth on the unseen family); replacing it with the breadth model makes the arm **inherit breadth’s tax** (−4.20 [−6.27, −2.14] against the clean-code teacher) while matching on seen conditions. Across all three teachers the student lands at roughly the teacher’s unseen-family level plus one point. The seen-condition gain comes from the supervised term and is teacher-independent; the unseen-family behaviour is *distilled from the teacher*. The method is “distil from a clean-code-*tuned* model on the clean parent”, and we name it as such.

What it repairs, and on which models. Table 16 reads all eight models from one evaluation phase; every previously published contrast recomputed through it falls inside the new interval (largest disagreement 0.85 points), so a departure is a property of the model. Breadth’s unseen-family tax has the same sign on every model, six significantly, across four lineages and 7B–34B.

Table 16. **RQ3: the three findings across eight models and four lineages**, all read from one evaluation phase. ✓ replicated, ○ inconclusive, × refuted. R1 is breadth’s unseen-family tax (mono_all1 – tuned_L0 on the held-out family); R2 is anchoring removing it (cons_lam3 – mono_all1); R3 is anchoring’s clean-code cost (cons_lam3 – tuned_L0 on clean code). Bold intervals exclude zero.

Model	Lineage	R1 unseen tax	R2 anchoring repair	R3 clean-code cost
CodeLlama-7B	Meta	−3.71 [−5.93, −1.56] ✓	+5.44 [+3.37, +7.41] ✓	−0.54 [−1.98, +1.08] ✓
CodeLlama-13B	Meta	−4.04 [−6.26, −1.89] ✓	+3.87 [+2.06, +5.77] ✓	−0.72 [−2.28, +0.90] ✓
CodeLlama-34B	Meta	−2.72 [−5.02, −0.49] ✓	+3.62 [+1.40, +5.76] ✓	+0.48 [−0.84, +1.80] ✓
Llama-3.1-8B	Meta	−2.72 [−4.70, −0.91] ✓	+3.38 [+1.57, +5.36] ✓	−0.30 [−1.98, +1.26] ✓
StarCoder2-15B	BigCode	−2.72 [−4.94, −0.58] ✓	+2.31 [+0.58, +4.03] ✓	−0.96 [−2.51, +0.60] ✓
Gemma-3-12B	Google	−4.70 [−6.93, −2.55] ✓	+2.64 [+0.91, +4.53] ✓	−1.80 [−3.53, −0.06] ×
CodeGemma-7B	Google	−1.98 [−3.95, +0.16] ○	−0.41 [−2.39, +1.49] ○	−2.04 [−3.77, −0.18] ×
Granite-3.1-8B	IBM	−0.91 [−2.97, +1.15] ○	−1.89 [−3.54, −0.25] ×	−3.41 [−5.39, −1.26] ×
<i>replicated / inconclusive / refuted</i>		6 / 2 / 0	6 / 1 / 1	5 / 0 / 3

The repair holds on six of eight models and **reverses on Granite-3.1-8B**, where the anchored arm is worse than breadth on the unseen family. “Anchoring pays no clean-code cost” holds on five of eight. Every failure falls outside the lineage the findings were discovered on, where all three are four-of-four; with $n = 8$ and vendor confounded with lineage (Fisher’s exact $p = 0.0714$ one-sided) that is a warning that single-lineage evaluation would have hidden all of it, not a claim about lineages.

Takeaway: RQ2

Distilling from a clean-code-*tuned* teacher on the unobfuscated parent, on exactly the rows breadth trains on, removes breadth’s unseen-family tax on six of eight models and reverses it on one (Granite-3.1-8B), at no clean-code cost on five of eight. The teacher ablation shows where the effect lives: the seen-condition gain is teacher-independent, the unseen-family behaviour is the teacher’s.

7 RQ3: Does the anchored model survive the same tests?

If the diagnosis is right, anchoring to something the transform cannot change should move both tests specifically where the input has left the training distribution.

Table 17. **RQ4: the divergence ladder**. Composites that contain a transform family the model has never seen. Breadth’s advantage flips sign the moment such a component enters, while anchoring rises *above* the clean-code control. The pre-registered rule required both that anchoring hold and that breadth fall below the control; only the first is established, since the pooled interval at d2 straddles zero. CodeLlama-7B, 287 programs common to all six composites, fixed before any system was evaluated.

Level	Stimulus	mono-clean	cons-clean	cons-mono
d0	depth 2, all seen	+3.49 [+2.04, +5.01]	−	+0.87 [−0.59, +2.33]
d1	depth 3–4, all seen	+4.15 [+2.37, +5.91]	+5.02 [+3.41, +6.69]	+0.87 [−0.59, +2.33]
d2	depth 2, unseen inside	−1.71 [−3.60, +0.31]	+2.13 [+0.66, +3.60]	+3.84 [+1.90, +5.70]
d3	depth 3, unseen inside	−0.35 [−2.67, +1.86]	+3.26 [+0.81, +5.92]	+3.60 [+1.40, +5.81]

Composition. Table 17 is the divergence ladder, from all-seen depth-2 stacks (d0) to stacks containing a family the model has never seen (d2, d3); the d2 and d3 stimulus did not exist before

this work. Once an unseen component enters, breadth’s advantage over the clean-code control **flips sign**, from +3.49 and +4.15 to −1.71 and −0.35, and anchoring, which merely matches breadth on seen stacks (+0.87 [−0.59, +2.33]), rises **above the control** (+2.13, +3.26) and **above breadth** (+3.84, +3.60). The pre-registered rule required both that anchoring hold and that breadth fall below the control; only the first is established, because the pooled d2 interval spans zero, so by that rule the outcome is UNDECIDED. Split by RQ1’s identifier rule, committed before the first d2 cell was written, identifier-containing stacks sit at +1.98 [−0.12, +4.18] and structural-containing stacks at −3.88 [−6.08, −1.71]: breadth is broken where the stack leaves it no identifier surface to key on. Across the panel (Table 34, Appendix D) the anchoring advantage on unseen-containing stacks replicates on five of eight models, is inconclusive on three and refuted on none; family exposure is significant on all eight and is the largest effect in the table.

Reversal. In Table 33, anchoring’s direction ratio is +0.03: the only arm in the supervised-fine-tuning family whose forward gain cost nothing backwards. It does **not** acquire backward competence (+0.62 [−1.56, +2.81] against the untuned model); the claim is that every existing method pays backward competence for its forward gain and anchoring is the first that does not. The one arm that gains in both directions is trained on the unseen family’s sibling (+1.63 [+0.11, +3.20] backwards): exposure to a family transfers in a way exposure to transforms does not.

The bound. On the unseen family alone the anchored model is as robust as clean-code tuning and no more (−0.30, −0.33, +0.74 against the control at 7B, 13B, 34B; seventh of thirty-three arms), and every arm that beats it there was trained on the family’s sibling. Anchoring buys breadth’s recombination benefit at little cost, extends it to composed inputs the model has never seen, and stops paying the backward price — and it does not add the capability breadth lacks. It is a repair of what adaptation breaks, not a route to invariance.

Takeaway: RQ3

The anchored model’s advantage appears only where the input diverges from training: level with breadth on seen stacks, above the clean-code control and above breadth once an unseen family enters (five of eight models replicate, none refute; the strict pre-registered rule is undecided), and a direction ratio of +0.03 where every other supervised arm is negative. On the unseen family alone it is as robust as clean-code tuning and no more: a repair of what adaptation breaks, not a route to invariance.

8 Related Work

This paper joins three distinct conversations in the literature. First, we differentiate from the deobfuscation (DOBF) lineage by strictly evaluating code that remains obfuscated, measuring semantic competence under transformation rather than recovery capability. Second, in the domain of model merging and modular adaptation, we contribute a negative result supported by a highly rigorous random-gate control, warning that task-vector geometry is often dominated by initialisation rather than genuine knowledge. Finally, we contribute to the growing literature on robustness to semantics-preserving transformations and direction specificity in code understanding.

9 Threats to Validity

The held-out family is a proxy. Every unseen-family number is measured on X1, a trainable sibling of the quarantined family H1, whose budget was spent under a pre-registered two-pass discipline; X1 reproduces H1’s difficulty at $r = 0.9992$ and transfer between them was lossless (0.08 points) for arms trained on the sibling. It is a single family pair; a second hard family with a different surface would settle whether “family” means “surface family”. **Coverage is not uniform.**

Conditions cover different program sets; every contrast runs on a subset recorded before evaluation, and where the intersection moves an estimate we report both. **Eight models is a small sample for a claim about models**, and Section 6 declines to make one; the panel's strongest non-lineage model is also the one an untuned capability screen would have discarded. **Coverage of the baselines**. Merging is measured forward only, on four models (Table 32); routing is measured on seen stacks on one model, so its verdict rests on matching breadth where the two are measured together; the direction ratio is one model deep, because the one-shot inverse template collapses the format on most other panel models. **Instrument failures are reported, not omitted**. The causal attention instrument is inert (an identifier knockout moves the gold log-probability by under 0.5 % everywhere), so the attention account is correlational; the weakest-link test behind the shared-surface reading in Section 5 returned undecided. **Single language**. Every headline is Python; the JavaScript corpus exists but cannot be regenerated on our cluster.

10 Data Availability

Data/code is available in project's website [].

References

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A Per-model absolute tables

The per-model absolute tables — every method against every obfuscation, output prediction and input prediction in one cell — are in the body, Section 4, one table per model for all eight panel models and the pilot (Tables 5–13).

B Per-model master grids

One table per panel model: every condition and every stack, forward and backward, with the raw accuracy, the change against the untuned model, and the percentage of the untuned model's clean-code accuracy. Cells marked *fmt* exceed the pre-registered format-failure gate (0.25) and enter no mean; the gate is applied per cell, which is stricter than the pooled registration. The untuned CodeLlama-34B sits at 0.254 on clean code, a hair over the bar, which gates its percentage column while its deltas stand. Rows marked '–' are cells not yet run at the time of generation.

C Routing and merging grids

Every routing and merging system on every condition and stack that has it, for the models on which those arms exist (routing: CodeLlama-7B; merging: CodeLlama-7B, Granite-3.1-8B, Llama-3.1-8B, StarCoder2-15B), with the same three numbers as Appendix B. The routing arms are MoLE mixtures of the eight per-transform experts; *uniform* and *random* gates separate the mixture's worth from the routing's. Every routing cell ran through the mixture engine.

D Supplementary tables

The contrast tables referenced from the research questions that are not the one table each question carries in the body: routing and merging on stacks (Table 31), merging across the panel (Table 32), the direction ratio (Table 33), the anchoring advantage by model (Table 34), and the cross-model summary in relative terms (Table 35).

Table 18. **Master grid, CodeLlama-7B.** Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.257	100	0.302	+4.5	117	0.427	+17.0	166	0.413	+15.6	161	0.422	+16.5	164	0.413	+15.6	161
L1b	0.198	77	0.255	+5.7	99	0.358	+16.0	139	0.388	+19.0	151	0.402	+20.4	156	0.346	+14.8	135
L1r	0.207	81	0.265	+5.8	103	0.378	+17.1	147	0.387	+18.0	151	0.395	+18.8	154	0.360	+15.3	140
L2	0.202	79	0.272	+7.0	106	0.382	+18.0	149	0.378	+17.7	147	0.398	+19.6	155	0.354	+15.2	138
S1	0.168	66	0.216	+4.7	84	0.379	+21.0	147	0.389	+22.1	151	0.400	+23.2	156	0.370	+20.2	144
S2	0.193	75	fmt	fmt	fmt	0.389	+19.6	152	0.403	+20.9	157	0.415	+22.2	162	0.383	+19.0	149
X1	0.119	46	-	-	-	0.269	+14.9	105	0.231	+11.2	90	0.286	+16.6	111	0.316	+19.7	123
C_L1b_S1	0.141	55	-	-	-	0.279	+13.7	108	0.345	+20.4	134	0.360	+21.9	140	0.278	+13.6	108
C_L1r_S1	0.131	51	-	-	-	0.294	+16.3	114	0.350	+21.9	136	0.356	+22.5	139	0.296	+16.5	115
C_S1_L1r	0.135	53	-	-	-	0.285	+15.0	111	0.326	+19.2	127	0.349	+21.4	136	0.297	+16.2	116
C_L2_S4	0.176	68	-	-	-	0.355	+17.9	138	0.385	+20.9	150	0.392	+21.7	153	0.343	+16.7	134
C_L1r_S3	0.172	67	-	-	-	0.360	+18.8	140	0.380	+20.8	148	0.387	+21.5	151	0.350	+17.8	136
C_S4_S3	0.186	72	-	-	-	0.392	+20.6	153	0.403	+21.7	157	0.422	+23.6	164	0.382	+19.6	149
seen stacks, pooled (6)	0.157	61	-	-	-	0.327	+17.1	127	0.365	+20.8	142	0.378	+22.1	147	0.324	+16.8	126
C3_L1r_S3_S4	0.150	58	-	-	-	0.340	+19.0	132	0.377	+22.7	147	0.380	+23.0	148	0.343	+19.3	133
C3_S1_S3_S4	0.128	50	-	-	-	0.354	+22.6	138	0.369	+24.1	144	0.385	+25.7	150	0.343	+21.5	133
C3_L1r_S1_S4	0.126	49	-	-	-	0.289	+16.3	112	0.353	+22.7	137	0.354	+22.7	138	0.303	+17.7	118
C4_L1r_S1_S3_S4	0.089	35	-	-	-	0.278	+18.8	108	0.336	+24.7	131	0.356	+26.7	139	0.292	+20.3	114
depth-3/4 stacks, pooled (4)	0.123	48	-	-	-	0.315	+19.2	123	0.359	+23.5	140	0.369	+24.5	144	0.320	+19.7	125
C_L1r_X1	0.102	40	-	-	-	0.184	+8.2	72	0.199	+9.6	77	0.222	+12.0	87	0.254	+15.2	99
C_X1_S1	0.104	40	-	-	-	0.274	+17.0	107	0.230	+12.6	89	0.267	+16.3	104	0.326	+22.2	127
C_S2_X1	0.145	57	-	-	-	0.313	+16.8	122	0.285	+13.9	111	0.332	+18.6	129	0.347	+20.2	135
unseen-containing stacks, pooled (3)	0.117	46	-	-	-	0.257	+14.0	100	0.238	+12.1	93	0.274	+15.7	107	0.309	+19.2	120
C_L1r_X1m	0.112	44	-	-	-	0.220	+10.7	85	0.232	+12.0	90	0.249	+13.7	97	0.281	+16.9	110
C_S1_X1s	0.151	59	-	-	-	0.334	+18.3	130	0.302	+15.1	117	0.355	+20.4	138	0.341	+19.0	133
C3_L1r_S1_X1	0.104	40	-	-	-	0.219	+11.5	85	0.222	+11.8	86	0.250	+14.6	97	0.271	+16.7	106
<i>Backward: input prediction, graded by execution</i>																	
L0	0.285	100	-	-	-	0.324	+3.9	114	0.276	-0.8	97	0.297	+1.3	104	0.312	+2.7	109
L1b	0.265	93	-	-	-	0.251	-1.3	88	0.279	+1.4	98	0.294	+2.9	103	0.265	+0.1	93
L1r	0.315	111	-	-	-	0.327	+1.1	115	0.276	-3.9	97	0.293	-2.3	103	0.327	+1.1	115
L2	0.282	99	-	-	-	0.324	+4.2	114	0.276	-0.6	97	0.275	-0.7	97	0.324	+4.2	114
S1	0.261	92	-	-	-	fmt	fmt	fmt	0.262	+0.1	92	0.280	+1.9	98	0.280	+1.8	98
S2	0.330	116	-	-	-	fmt	fmt	fmt	0.294	-3.6	103	0.332	+0.2	117	0.330	+0.0	116
X1	0.257	90	-	-	-	fmt	fmt	fmt	0.237	-2.0	83	0.281	+2.5	99	0.291	+3.4	102
C_L1b_S1	0.218	76	-	-	-	fmt	fmt	fmt	0.242	+2.5	85	0.270	+5.3	95	0.246	+2.8	86
C_L1r_S1	0.301	106	-	-	-	fmt	fmt	fmt	0.255	-4.6	90	0.270	-3.1	95	0.300	-0.2	105
C_S1_L1r	0.282	99	-	-	-	0.205	-7.7	72	0.267	-1.5	94	0.285	+0.2	100	0.262	-2.0	92
C_L2_S4	0.291	102	-	-	-	fmt	fmt	fmt	0.260	-3.1	91	0.283	-0.8	99	0.315	+2.4	111
C_L1r_S3	0.296	104	-	-	-	0.267	-2.9	94	0.278	-1.9	97	0.315	+1.9	111	0.330	+3.4	116
C_S4_S3	0.315	111	-	-	-	fmt	fmt	fmt	0.282	-3.4	99	0.333	+1.7	117	0.324	+0.8	114
seen stacks, pooled (6)	0.284	100	-	-	-	fmt	fmt	fmt	0.264	-2.0	93	0.293	+0.9	103	0.296	+1.2	104
C3_L1r_S3_S4	0.303	106	-	-	-	fmt	fmt	fmt	0.268	-3.5	94	0.302	-0.1	106	0.306	+0.4	108
C3_S1_S3_S4	0.292	103	-	-	-	fmt	fmt	fmt	0.234	-5.8	82	0.313	+2.1	110	0.291	-0.1	102
C3_L1r_S1_S4	0.299	105	-	-	-	fmt	fmt	fmt	0.229	-7.0	80	0.278	-2.2	97	0.262	-3.7	92
C4_L1r_S1_S3_S4	0.282	99	-	-	-	fmt	fmt	fmt	0.217	-6.5	76	0.288	+0.6	101	0.273	-1.0	96
depth-3/4 stacks, pooled (4)	0.294	103	-	-	-	fmt	fmt	fmt	0.237	-5.7	83	0.295	+0.1	104	0.283	-1.1	99
C_L1r_X1	0.253	89	-	-	-	0.210	-4.3	74	0.200	-5.3	70	0.250	-0.2	88	0.245	-0.7	86
C_X1_S1	0.251	88	-	-	-	fmt	fmt	fmt	0.201	-5.0	71	0.276	+2.5	97	0.255	+0.4	90
C_S2_X1	0.311	109	-	-	-	fmt	fmt	fmt	0.217	-9.4	76	0.305	-0.5	107	0.280	-3.0	98
unseen-containing stacks, pooled (3)	0.271	95	-	-	-	fmt	fmt	fmt	0.206	-6.5	72	0.277	+0.6	97	0.260	-1.1	91
C_L1r_X1m	0.250	88	-	-	-	0.220	-2.9	77	0.223	-2.7	78	0.250	+0.0	88	0.233	-1.7	82
C_S1_X1s	0.277	97	-	-	-	fmt	fmt	fmt	0.195	-8.1	69	0.288	+1.1	101	0.270	-0.6	95
C3_L1r_S1_X1	0.263	92	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	0.231	-3.2	81	0.254	-0.9	89

Table 19. **Master grid, CodeLlama-13B.** Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.252	100	fmt	fmt	fmt	0.469	+21.7	186	0.446	+19.3	177	0.462	+21.0	183	0.463	+21.1	184
L1b	0.222	88	fmt	fmt	fmt	0.397	+17.6	158	0.423	+20.1	168	0.438	+21.7	174	0.379	+15.7	150
L1r	0.225	89	fmt	fmt	fmt	0.415	+19.0	165	0.420	+19.5	167	0.439	+21.4	174	0.407	+18.1	161
L2	0.228	90	fmt	fmt	fmt	0.407	+17.8	161	0.417	+18.9	166	0.432	+20.4	171	0.401	+17.3	159
S1	0.213	85	0.216	+0.2	86	0.408	+19.5	162	0.413	+20.0	164	0.435	+22.1	172	0.410	+19.6	163
S2	0.184	73	fmt	fmt	fmt	0.422	+23.8	168	0.445	+26.0	176	0.469	+28.4	186	0.422	+23.8	168
X1	0.149	59	-	-	-	0.294	+14.5	117	0.254	+10.5	101	0.292	+14.3	116	0.353	+20.4	140
C_L1b_S1	0.183	72	-	-	-	0.318	+13.6	126	0.367	+18.4	146	0.375	+19.2	149	0.320	+13.7	127
C_L1r_S1	0.175	69	-	-	-	0.318	+14.4	126	0.371	+19.6	147	0.376	+20.1	149	0.342	+16.8	136
C_S1_L1r	0.172	68	-	-	-	0.315	+14.4	125	0.359	+18.8	142	0.363	+19.2	144	0.350	+17.8	139
C_L2_S4	0.200	79	-	-	-	0.389	+19.0	154	0.422	+22.2	167	0.432	+23.2	171	0.385	+18.5	153
C_L1r_S3	0.186	74	-	-	-	0.407	+22.0	161	0.409	+22.3	162	0.434	+24.8	172	0.399	+21.3	158
C_S4_S3	0.192	76	-	-	-	0.424	+23.2	168	0.436	+24.4	173	0.459	+26.7	182	0.433	+24.1	172
seen stacks, pooled (6)	0.185	73	-	-	-	0.362	+17.8	144	0.394	+21.0	156	0.407	+22.2	161	0.372	+18.7	147
C3_L1r_S3_S4	0.166	66	-	-	-	0.382	+21.6	152	0.410	+24.4	163	0.429	+26.3	170	0.382	+21.5	151
C3_S1_S3_S4	0.147	58	-	-	-	0.367	+22.0	146	0.408	+26.1	162	0.423	+27.6	168	0.384	+23.7	152
C3_L1r_S1_S4	0.163	65	-	-	-	0.314	+15.1	124	0.367	+20.4	146	0.367	+20.4	146	0.349	+18.6	138
C4_L1r_S1_S3_S4	0.117	46	-	-	-	0.300	+18.4	119	0.353	+23.6	140	0.360	+24.3	143	0.331	+21.5	131
depth-3/4 stacks, pooled (4)	0.148	59	-	-	-	0.341	+19.3	135	0.384	+23.6	152	0.395	+24.7	157	0.361	+21.3	143
C_L1r_X1	0.123	49	-	-	-	0.199	+7.7	79	0.213	+9.0	84	0.236	+11.3	93	0.284	+16.1	113
C_X1_S1	0.138	55	-	-	-	0.295	+15.7	117	0.270	+13.2	107	0.307	+16.8	122	0.341	+20.3	135
C_S2_X1	0.160	64	-	-	-	0.335	+17.4	133	0.334	+17.4	133	0.356	+19.6	141	0.376	+21.6	149
unseen-containing stacks, pooled (3)	0.140	56	-	-	-	0.276	+13.6	110	0.272	+13.2	108	0.300	+15.9	119	0.334	+19.3	132
C_L1r_X1m	0.138	55	-	-	-	0.246	+10.8	98	0.248	+11.0	98	0.279	+14.1	110	0.319	+18.2	127
C_S1_X1s	0.179	71	-	-	-	0.364	+18.5	144	0.345	+16.6	137	0.360	+18.1	143	0.368	+18.9	146
C3_L1r_S1_X1	0.128	51	-	-	-	0.233	+10.5	92	0.234	+10.6	93	0.249	+12.1	99	0.291	+16.3	116
<i>Backward: input prediction, graded by execution</i>																	
L0	0.285	100	-	-	-	0.269	-1.6	94	fmt	fmt	fmt	0.231	-5.4	81	fmt	fmt	fmt
L1b	0.277	97	-	-	-	0.285	+0.8	100	fmt	fmt	fmt	0.253	-2.5	89	fmt	fmt	fmt
L1r	0.327	115	-	-	-	0.347	+2.0	122	fmt	fmt	fmt	0.229	-9.8	80	fmt	fmt	fmt
L2	0.279	98	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	0.206	-7.2	72	fmt	fmt	fmt
S1	0.304	107	-	-	-	0.341	+3.7	120	fmt	fmt	fmt	0.199	-10.6	70	fmt	fmt	fmt
S2	0.293	103	-	-	-	0.317	+2.4	111	fmt	fmt	fmt	0.238	-5.5	84	fmt	fmt	fmt
X1	0.258	91	-	-	-	0.274	+1.6	96	fmt	fmt	fmt	0.219	-4.0	77	fmt	fmt	fmt
C_L1b_S1	0.253	89	-	-	-	0.326	+7.3	114	fmt	fmt	fmt	0.201	-5.2	71	fmt	fmt	fmt
C_L1r_S1	0.325	114	-	-	-	0.360	+3.5	126	fmt	fmt	fmt	0.203	-12.1	71	fmt	fmt	fmt
C_S1_L1r	0.323	113	-	-	-	0.351	+2.8	123	fmt	fmt	fmt	0.230	-9.3	81	fmt	fmt	fmt
C_L2_S4	0.274	96	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_S3	0.305	107	-	-	-	0.361	+5.6	127	fmt	fmt	fmt	0.220	-8.5	77	fmt	fmt	fmt
C_S4_S3	0.293	103	-	-	-	0.326	+3.4	114	fmt	fmt	fmt	0.239	-5.3	84	fmt	fmt	fmt
seen stacks, pooled (6)	0.295	104	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C3_L1r_S3_S4	0.310	109	-	-	-	0.337	+2.7	118	fmt	fmt	fmt	0.224	-8.6	79	fmt	fmt	fmt
C3_S1_S3_S4	0.295	103	-	-	-	0.329	+3.5	116	fmt	fmt	fmt	0.235	-5.9	83	fmt	fmt	fmt
C3_L1r_S1_S4	0.316	111	-	-	-	0.353	+3.7	124	fmt	fmt	fmt	0.211	-10.5	74	fmt	fmt	fmt
C4_L1r_S1_S3_S4	0.310	109	-	-	-	0.333	+2.3	117	fmt	fmt	fmt	0.222	-8.9	78	fmt	fmt	fmt
depth-3/4 stacks, pooled (4)	0.308	108	-	-	-	0.338	+3.1	119	fmt	fmt	fmt	0.223	-8.5	78	fmt	fmt	fmt
C_L1r_X1	0.281	99	-	-	-	0.254	-2.6	89	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_X1_S1	0.289	101	-	-	-	0.264	-2.5	93	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_S2_X1	0.271	95	-	-	-	0.279	+0.8	98	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
unseen-containing stacks, pooled (3)	0.280	98	-	-	-	0.266	-1.4	93	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_X1m	0.291	102	-	-	-	0.259	-3.1	91	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_S1_X1s	0.301	106	-	-	-	0.290	-1.1	102	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C3_L1r_S1_X1	0.304	107	-	-	-	0.293	-1.0	103	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt

Table 20. **Master grid, CodeLlama-34B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base	acc	base_1shot	acc	Δ	tuned_L0	acc	Δ	mono_all	acc	Δ	cons_lam3	acc	Δ	tuned_X1	acc	Δ
<i>Forward: output prediction</i>																	
L0	fmt	fmt	fmt	fmt	fmt	0.520	fmt	fmt	0.494	fmt	fmt	0.525	fmt	fmt	0.502	fmt	fmt
L1b	0.205	fmt	fmt	fmt	fmt	0.417	+21.2	fmt	0.469	+26.4	fmt	0.494	+28.9	fmt	0.401	+19.6	fmt
L1r	0.220	fmt	fmt	fmt	fmt	0.469	+25.0	fmt	0.457	+23.8	fmt	0.487	+26.8	fmt	0.459	+24.0	fmt
L2	0.231	fmt	fmt	fmt	fmt	0.466	+23.5	fmt	0.465	+23.5	fmt	0.478	+24.7	fmt	0.456	+22.6	fmt
S1	0.222	fmt	fmt	fmt	fmt	0.472	+24.9	fmt	0.467	+24.5	fmt	0.496	+27.3	fmt	0.461	+23.9	fmt
S2	0.191	fmt	fmt	fmt	fmt	0.484	+29.3	fmt	0.484	+29.3	fmt	0.525	+33.5	fmt	0.478	+28.7	fmt
X1	0.135	fmt	-	-	-	0.325	+19.0	fmt	0.298	+16.3	fmt	0.334	+19.9	fmt	0.381	+24.5	fmt
C_L1b_S1	0.156	fmt	-	-	-	0.322	+16.6	fmt	0.399	+24.3	fmt	0.421	+26.5	fmt	0.336	+18.0	fmt
C_L1r_S1	0.184	fmt	-	-	-	0.351	+16.8	fmt	0.393	+21.0	fmt	0.427	+24.3	fmt	0.366	+18.2	fmt
C_S1_L1r	0.180	fmt	-	-	-	0.367	+18.8	fmt	0.405	+22.5	fmt	0.425	+24.6	fmt	0.379	+20.0	fmt
C_L2_S4	0.191	fmt	-	-	-	0.431	+24.1	fmt	0.454	+26.3	fmt	0.478	+28.7	fmt	0.431	+24.1	fmt
C_L1r_S3	0.169	fmt	-	-	-	0.451	+28.1	fmt	0.454	+28.5	fmt	0.482	+31.3	fmt	0.437	+26.8	fmt
C_S4_S3	0.199	fmt	-	-	-	0.482	+28.3	fmt	0.487	+28.7	fmt	0.521	+32.2	fmt	0.476	+27.7	fmt
<i>seen stacks, pooled (6)</i>	0.180	fmt	-	-	-	0.401	+22.1	fmt	0.432	+25.2	fmt	0.459	+27.9	fmt	0.404	+22.4	fmt
C3_L1r_S3_S4	0.148	fmt	-	-	-	0.416	+26.8	fmt	0.452	+30.4	fmt	0.473	+32.5	fmt	0.424	+27.5	fmt
C3_S1_S3_S4	0.128	fmt	-	-	-	0.428	+30.1	fmt	0.454	+32.7	fmt	0.472	+34.5	fmt	0.440	+31.2	fmt
C3_L1r_S1_S4	0.147	fmt	-	-	-	0.338	+19.2	fmt	0.390	+24.3	fmt	0.412	+26.5	fmt	0.359	+21.2	fmt
C4_L1r_S1_S3_S4	0.105	fmt	-	-	-	0.316	+21.1	fmt	0.372	+26.7	fmt	0.413	+30.8	fmt	0.354	+24.9	fmt
<i>depth-3/4 stacks, pooled (4)</i>	0.132	fmt	-	-	-	0.375	+24.3	fmt	0.417	+28.5	fmt	0.442	+31.1	fmt	0.394	+26.2	fmt
C_L1r_X1	0.115	fmt	-	-	-	0.226	+11.0	fmt	0.249	+13.3	fmt	0.278	+16.2	fmt	0.275	+16.0	fmt
C_X1_S1	0.125	fmt	-	-	-	0.317	+19.2	fmt	0.290	+16.5	fmt	0.339	+21.4	fmt	0.372	+24.7	fmt
C_S2_X1	0.176	fmt	-	-	-	0.372	+19.6	fmt	0.355	+17.9	fmt	0.403	+22.7	fmt	0.395	+21.9	fmt
<i>unseen-containing stacks, pooled (3)</i>	0.139	fmt	-	-	-	0.305	+16.6	fmt	0.298	+15.9	fmt	0.340	+20.1	fmt	0.348	+20.9	fmt
C_L1r_X1m	0.128	fmt	-	-	-	0.261	+13.3	fmt	0.275	+14.6	fmt	0.294	+16.5	fmt	0.317	+18.9	fmt
C_S1_X1s	0.176	fmt	-	-	-	0.390	+21.4	fmt	0.382	+20.6	fmt	0.415	+23.9	fmt	0.411	+23.5	fmt
C3_L1r_S1_X1	0.118	fmt	-	-	-	0.256	+13.8	fmt	0.262	+14.4	fmt	0.297	+17.9	fmt	0.293	+17.5	fmt
<i>Backward: input prediction, graded by execution</i>																	
L0	0.363	100	-	-	-	0.260	-10.3	72	fmt	fmt	fmt	0.284	-7.8	78	0.269	-9.4	74
L1b	0.327	90	-	-	-	0.253	-7.4	70	fmt	fmt	fmt	0.287	-4.0	79	0.227	-10.0	63
L1r	0.339	93	-	-	-	0.255	-8.4	70	fmt	fmt	fmt	0.256	-8.3	71	0.241	-9.8	67
L2	0.357	99	-	-	-	0.280	-7.7	77	fmt	fmt	fmt	0.276	-8.1	76	0.268	-8.9	74
S1	0.323	89	-	-	-	0.305	-1.8	84	fmt	fmt	fmt	0.264	-5.8	73	0.246	-7.7	68
S2	0.350	96	-	-	-	0.354	+0.4	98	fmt	fmt	fmt	0.306	-4.4	84	0.278	-7.2	77
X1	0.303	84	-	-	-	0.253	-5.0	70	fmt	fmt	fmt	0.249	-5.3	69	0.207	-9.6	57
C_L1b_S1	0.257	71	-	-	-	0.272	+1.5	75	fmt	fmt	fmt	0.258	+0.1	71	fmt	fmt	fmt
C_L1r_S1	0.304	84	-	-	-	0.285	-1.8	79	fmt	fmt	fmt	0.200	-10.4	55	0.254	-4.9	70
C_S1_L1r	0.292	80	-	-	-	0.307	+1.5	85	fmt	fmt	fmt	0.216	-7.6	60	0.264	-2.8	73
C_L2_S4	0.344	95	-	-	-	0.291	-5.3	80	fmt	fmt	fmt	0.273	-7.1	75	0.258	-8.6	71
C_L1r_S3	0.333	92	-	-	-	0.327	-0.7	90	fmt	fmt	fmt	0.256	-7.7	71	0.269	-6.5	74
C_S4_S3	0.352	97	-	-	-	0.345	-0.7	95	fmt	fmt	fmt	0.301	-5.1	83	0.276	-7.6	76
<i>seen stacks, pooled (6)</i>	0.314	86	-	-	-	0.305	-0.9	84	fmt	fmt	fmt	0.251	-6.3	69	fmt	fmt	fmt
C3_L1r_S3_S4	0.325	90	-	-	-	0.330	+0.5	91	fmt	fmt	fmt	0.251	-7.4	69	0.272	-5.3	75
C3_S1_S3_S4	0.330	91	-	-	-	0.323	-0.7	89	fmt	fmt	fmt	0.288	-4.2	79	0.270	-6.0	74
C3_L1r_S1_S4	0.291	80	-	-	-	0.311	+2.0	86	fmt	fmt	fmt	0.199	-9.2	55	0.271	-2.0	75
C4_L1r_S1_S3_S4	0.283	78	-	-	-	0.333	+5.0	92	fmt	fmt	fmt	0.214	-6.9	59	0.270	-1.3	75
<i>depth-3/4 stacks, pooled (4)</i>	0.307	85	-	-	-	0.324	+1.7	89	fmt	fmt	fmt	0.238	-6.9	66	0.271	-3.6	75
C_L1r_X1	0.275	76	-	-	-	0.240	-3.5	66	fmt	fmt	fmt	fmt	fmt	fmt	0.195	-8.0	54
C_X1_S1	0.298	82	-	-	-	0.275	-2.3	76	fmt	fmt	fmt	fmt	fmt	fmt	0.228	-7.0	63
C_S2_X1	0.378	104	-	-	-	0.366	-1.3	101	fmt	fmt	fmt	0.288	-9.1	79	0.233	-14.5	64
<i>unseen-containing stacks, pooled (3)</i>	0.317	87	-	-	-	0.294	-2.3	81	fmt	fmt	fmt	fmt	fmt	fmt	0.219	-9.8	60
C_L1r_X1m	0.269	74	-	-	-	0.230	-3.9	63	fmt	fmt	fmt	fmt	fmt	fmt	0.189	-8.0	52
C_S1_X1s	0.346	95	-	-	-	0.302	-4.4	83	fmt	fmt	fmt	0.268	-7.8	74	0.250	-9.6	69
C3_L1r_S1_X1	0.304	84	-	-	-	0.296	-0.8	82	fmt	fmt	fmt	fmt	fmt	fmt	0.237	-6.6	65

Table 21. **Master grid, Llama-3.1-8B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.258	100	0.320	+6.2	124	0.448	+19.0	174	0.426	+16.8	165	0.445	+18.7	172	0.459	+20.1	178
L1b	0.222	86	0.276	+5.4	107	0.361	+13.9	140	0.387	+16.5	150	0.407	+18.5	158	0.369	+14.7	143
L1r	0.242	94	0.284	+4.3	110	0.387	+14.6	150	0.387	+14.5	150	0.405	+16.3	157	0.378	+13.7	147
L2	0.215	83	0.278	+6.3	108	0.391	+17.6	152	0.386	+17.1	150	0.409	+19.4	158	0.387	+17.2	150
S1	0.213	83	0.256	+4.3	99	0.395	+18.2	153	0.391	+17.7	151	0.413	+20.0	160	0.409	+19.6	158
S2	0.205	79	0.272	+6.7	106	0.413	+20.8	160	0.413	+20.8	160	0.445	+23.9	172	0.438	+23.3	170
X1	0.132	51	-	-	-	0.249	+11.7	96	0.222	+9.0	86	0.255	+12.4	99	0.332	+20.0	129
C_L1b_S1	0.152	59	-	-	-	0.279	+12.7	108	0.338	+18.7	131	0.368	+21.6	143	0.305	+15.3	118
C_L1r_S1	0.162	63	-	-	-	0.303	+14.1	118	0.352	+19.0	136	0.363	+20.1	141	0.314	+15.2	122
C_S1_L1r	0.159	62	-	-	-	0.277	+11.8	107	0.338	+17.9	131	0.354	+19.5	137	0.305	+14.6	118
C_L2_S4	0.205	79	-	-	-	0.349	+14.3	135	0.386	+18.1	149	0.401	+19.6	155	0.350	+14.5	136
C_L1r_S3	0.198	77	-	-	-	0.367	+16.9	142	0.382	+18.4	148	0.398	+19.9	154	0.369	+17.1	143
C_S4_S3	0.214	83	-	-	-	0.421	+20.7	163	0.411	+19.7	159	0.444	+23.0	172	0.431	+21.8	167
<i>seen stacks, pooled (6)</i>	0.182	70	-	-	-	0.332	+15.1	129	0.368	+18.6	142	0.388	+20.6	150	0.346	+16.4	134
C3_L1r_S3_S4	0.183	71	-	-	-	0.329	+14.6	128	0.374	+19.1	145	0.391	+20.8	152	0.363	+18.0	141
C3_S1_S3_S4	0.160	62	-	-	-	0.359	+19.8	139	0.376	+21.5	146	0.393	+23.2	152	0.378	+21.8	147
C3_L1r_S1_S4	0.160	62	-	-	-	0.279	+11.8	108	0.342	+18.2	133	0.358	+19.7	139	0.299	+13.9	116
C4_L1r_S1_S3_S4	0.140	54	-	-	-	0.263	+12.4	102	0.335	+19.6	130	0.358	+21.8	139	0.294	+15.4	114
<i>depth-3/4 stacks, pooled (4)</i>	0.161	62	-	-	-	0.308	+14.7	119	0.357	+19.6	138	0.375	+21.4	145	0.334	+17.3	129
C_L1r_X1	0.100	39	-	-	-	0.165	+6.4	64	0.181	+8.1	70	0.207	+10.6	80	0.257	+15.7	100
C_X1_S1	0.130	50	-	-	-	0.271	+14.1	105	0.225	+9.5	87	0.253	+12.3	98	0.300	+17.0	116
C_S2_X1	0.170	66	-	-	-	0.296	+12.6	115	0.284	+11.4	110	0.317	+14.7	123	0.360	+19.0	139
<i>unseen-containing stacks, pooled (3)</i>	0.134	52	-	-	-	0.244	+11.1	95	0.230	+9.6	89	0.259	+12.5	100	0.306	+17.2	118
C_L1r_X1m	0.115	45	-	-	-	0.184	+6.9	71	0.216	+10.1	84	0.232	+11.7	90	0.291	+17.6	113
C_S1_X1s	0.193	75	-	-	-	0.341	+14.8	132	0.299	+10.6	116	0.335	+14.2	130	0.355	+16.2	138
C3_L1r_S1_X1	0.105	41	-	-	-	0.194	+8.9	75	0.207	+10.2	80	0.224	+11.9	87	0.251	+14.6	97
<i>Backward: input prediction, graded by execution</i>																	
L0	0.409	100	-	-	-	0.335	-7.4	82	0.320	-9.0	78	0.332	-7.8	81	0.365	-4.4	89
L1b	0.371	91	-	-	-	0.297	-7.4	73	0.280	-9.2	68	0.285	-8.6	70	0.297	-7.5	72
L1r	0.373	91	-	-	-	0.312	-6.1	76	0.290	-8.3	71	0.297	-7.5	73	0.316	-5.7	77
L2	0.376	92	-	-	-	0.297	-7.8	73	0.284	-9.2	69	0.294	-8.1	72	0.319	-5.7	78
S1	0.375	92	-	-	-	0.325	-5.0	79	0.285	-9.0	70	0.291	-8.4	71	0.360	-1.5	88
S2	0.378	92	-	-	-	0.345	-3.4	84	0.326	-5.2	80	0.325	-5.3	79	0.366	-1.3	89
X1	0.263	64	-	-	-	0.267	+0.3	65	0.211	-5.3	51	0.228	-3.5	56	0.302	+3.9	74
C_L1b_S1	0.287	70	-	-	-	0.253	-3.4	62	0.274	-1.4	67	0.258	-2.9	63	0.297	+1.0	72
C_L1r_S1	0.297	73	-	-	-	0.281	-1.7	69	0.280	-1.8	68	0.283	-1.4	69	0.271	-2.6	66
C_S1_L1r	0.303	74	-	-	-	0.269	-3.4	66	0.274	-2.9	67	0.274	-2.9	67	0.290	-1.3	71
C_L2_S4	0.320	78	-	-	-	0.275	-4.6	67	0.269	-5.2	66	0.286	-3.4	70	0.309	-1.1	75
C_L1r_S3	0.342	84	-	-	-	0.297	-4.5	73	0.299	-4.3	73	0.303	-3.9	74	0.308	-3.4	75
C_S4_S3	0.367	90	-	-	-	0.333	-3.4	81	0.313	-5.4	76	0.329	-3.8	80	0.362	-0.5	88
<i>seen stacks, pooled (6)</i>	0.319	78	-	-	-	0.284	-3.5	70	0.285	-3.5	70	0.289	-3.1	71	0.306	-1.3	75
C3_L1r_S3_S4	0.309	75	-	-	-	0.271	-3.8	66	0.303	-0.5	74	0.300	-0.8	73	0.300	-0.9	73
C3_S1_S3_S4	0.353	86	-	-	-	0.311	-4.2	76	0.289	-6.3	71	0.281	-7.2	69	0.340	-1.3	83
C3_L1r_S1_S4	0.302	74	-	-	-	0.270	-3.2	66	0.262	-4.0	64	0.271	-3.1	66	0.270	-3.2	66
C4_L1r_S1_S3_S4	0.281	69	-	-	-	0.234	-4.7	57	0.280	-0.2	68	0.279	-0.2	68	0.259	-2.2	63
<i>depth-3/4 stacks, pooled (4)</i>	0.311	76	-	-	-	0.272	-4.0	66	0.284	-2.8	69	0.283	-2.8	69	0.292	-1.9	71
C_L1r_X1	0.218	53	-	-	-	0.220	+0.2	54	0.193	-2.6	47	0.239	+2.1	58	0.265	+4.7	65
C_X1_S1	0.291	71	-	-	-	0.283	-0.7	69	0.210	-8.1	51	0.220	-7.1	54	0.302	+1.1	74
C_S2_X1	0.323	79	-	-	-	0.288	-3.4	70	0.229	-9.3	56	0.241	-8.1	59	0.306	-1.6	75
<i>unseen-containing stacks, pooled (3)</i>	0.277	68	-	-	-	0.264	-1.3	64	0.211	-6.7	51	0.233	-4.4	57	0.291	+1.4	71
C_L1r_X1m	0.232	57	-	-	-	0.210	-2.2	51	0.217	-1.5	53	0.245	+1.3	60	0.258	+2.7	63
C_S1_X1s	0.357	87	-	-	-	0.287	-7.0	70	0.205	-15.2	50	0.212	-14.5	52	0.330	-2.7	81
C3_L1r_S1_X1	0.268	65	-	-	-	0.234	-3.4	57	0.188	-8.0	46	0.194	-7.4	47	0.243	-2.5	59

Table 22. **Master grid, StarCoder2-15B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base	acc	base	acc	%	tuned_L0	acc	%	mono_all	acc	%	cons_lam3	acc	%	tuned_X1	acc	%
<i>Forward: output prediction</i>																	
L0	fnt	fnt	fnt	fnt	fnt	0.542	fnt	fnt	0.529	fnt	fnt	0.532	fnt	fnt	0.513	fnt	fnt
L1b	fnt	fnt	fnt	fnt	fnt	0.431	fnt	fnt	0.487	fnt	fnt	0.510	fnt	fnt	0.402	fnt	fnt
L1r	fnt	fnt	fnt	fnt	fnt	0.483	fnt	fnt	0.485	fnt	fnt	0.509	fnt	fnt	0.448	fnt	fnt
L2	fnt	fnt	fnt	fnt	fnt	0.479	fnt	fnt	0.486	fnt	fnt	0.503	fnt	fnt	0.437	fnt	fnt
S1	fnt	fnt	fnt	fnt	fnt	0.508	fnt	fnt	0.487	fnt	fnt	0.501	fnt	fnt	0.470	fnt	fnt
S2	fnt	fnt	fnt	fnt	fnt	0.512	fnt	fnt	0.514	fnt	fnt	0.539	fnt	fnt	0.501	fnt	fnt
X1	fnt	fnt	-	-	-	0.347	fnt	fnt	0.320	fnt	fnt	0.343	fnt	fnt	0.381	fnt	fnt
C_L1b_S1	fnt	fnt	-	-	-	0.360	fnt	fnt	0.416	fnt	fnt	0.456	fnt	fnt	0.336	fnt	fnt
C_L1r_S1	fnt	fnt	-	-	-	0.393	fnt	fnt	0.423	fnt	fnt	0.440	fnt	fnt	0.377	fnt	fnt
C_S1_L1r	fnt	fnt	-	-	-	0.377	fnt	fnt	0.406	fnt	fnt	0.431	fnt	fnt	0.370	fnt	fnt
C_L2_S4	fnt	fnt	-	-	-	0.454	fnt	fnt	0.470	fnt	fnt	0.508	fnt	fnt	0.421	fnt	fnt
C_L1r_S3	fnt	fnt	-	-	-	0.473	fnt	fnt	0.483	fnt	fnt	0.504	fnt	fnt	0.435	fnt	fnt
C_S4_S3	fnt	fnt	-	-	-	0.511	fnt	fnt	0.516	fnt	fnt	0.539	fnt	fnt	0.490	fnt	fnt
seen stacks, pooled (6)	fnt	fnt	-	-	-	0.428	fnt	fnt	0.452	fnt	fnt	0.480	fnt	fnt	0.405	fnt	fnt
C3_L1r_S3_S4	fnt	fnt	-	-	-	0.448	fnt	fnt	0.484	fnt	fnt	0.503	fnt	fnt	0.413	fnt	fnt
C3_S1_S3_S4	fnt	fnt	-	-	-	0.476	fnt	fnt	0.465	fnt	fnt	0.493	fnt	fnt	0.459	fnt	fnt
C3_L1r_S1_S4	fnt	fnt	-	-	-	0.370	fnt	fnt	0.417	fnt	fnt	0.439	fnt	fnt	0.366	fnt	fnt
C4_L1r_S1_S3_S4	fnt	fnt	-	-	-	0.357	fnt	fnt	0.412	fnt	fnt	0.437	fnt	fnt	0.353	fnt	fnt
depth-3/4 stacks, pooled (4)	fnt	fnt	-	-	-	0.412	fnt	fnt	0.444	fnt	fnt	0.468	fnt	fnt	0.398	fnt	fnt
C_L1r_X1	fnt	fnt	-	-	-	0.248	fnt	fnt	0.274	fnt	fnt	0.286	fnt	fnt	0.288	fnt	fnt
C_X1_S1	fnt	fnt	-	-	-	0.357	fnt	fnt	0.322	fnt	fnt	0.342	fnt	fnt	0.401	fnt	fnt
C_S2_X1	fnt	fnt	-	-	-	0.403	fnt	fnt	0.375	fnt	fnt	0.397	fnt	fnt	0.420	fnt	fnt
unseen-containing stacks, pooled (3)	fnt	fnt	-	-	-	0.336	fnt	fnt	0.324	fnt	fnt	0.341	fnt	fnt	0.370	fnt	fnt
C_L1r_X1m	fnt	fnt	-	-	-	0.307	fnt	fnt	0.323	fnt	fnt	0.338	fnt	fnt	0.342	fnt	fnt
C_S1_X1s	fnt	fnt	-	-	-	0.434	fnt	fnt	0.408	fnt	fnt	0.434	fnt	fnt	0.435	fnt	fnt
C3_L1r_S1_X1	fnt	fnt	-	-	-	0.278	fnt	fnt	0.298	fnt	fnt	0.302	fnt	fnt	0.306	fnt	fnt
<i>Backward: input prediction, graded by execution</i>																	
L0	0.366	100	-	-	-	0.411	+4.4	112	0.311	-5.6	85	0.399	+3.2	109	0.393	+2.7	107
L1b	fnt	fnt	-	-	-	0.316	fnt	86	0.301	fnt	82	0.395	fnt	108	fnt	fnt	fnt
L1r	fnt	fnt	-	-	-	0.354	fnt	97	0.273	fnt	75	0.369	fnt	101	0.298	fnt	81
L2	0.355	97	-	-	-	0.391	+3.7	107	0.297	-5.7	81	0.379	+2.4	103	0.363	+0.8	99
S1	fnt	fnt	-	-	-	0.361	fnt	98	fnt	fnt	fnt	0.364	fnt	99	0.342	fnt	93
S2	fnt	fnt	-	-	-	0.359	fnt	98	0.300	fnt	82	0.384	fnt	105	fnt	fnt	fnt
X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.281	fnt	77	fnt	fnt	fnt
C_L1b_S1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.317	fnt	87	fnt	fnt	fnt
C_L1r_S1	fnt	fnt	-	-	-	0.301	fnt	82	fnt	fnt	fnt	0.303	fnt	83	fnt	fnt	fnt
C_S1_L1r	fnt	fnt	-	-	-	0.311	fnt	85	fnt	fnt	fnt	0.303	fnt	83	fnt	fnt	fnt
C_L2_S4	fnt	fnt	-	-	-	0.342	fnt	93	fnt	fnt	fnt	0.353	fnt	96	0.290	fnt	79
C_L1r_S3	fnt	fnt	-	-	-	0.300	fnt	82	0.282	fnt	77	0.368	fnt	100	fnt	fnt	fnt
C_S4_S3	fnt	fnt	-	-	-	0.368	fnt	101	0.301	fnt	82	0.386	fnt	105	fnt	fnt	fnt
seen stacks, pooled (6)	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.338	fnt	92	fnt	fnt	fnt
C3_L1r_S3_S4	fnt	fnt	-	-	-	0.294	fnt	80	fnt	fnt	fnt	0.339	fnt	92	fnt	fnt	fnt
C3_S1_S3_S4	fnt	fnt	-	-	-	0.337	fnt	92	fnt	fnt	fnt	0.333	fnt	91	fnt	fnt	fnt
C3_L1r_S1_S4	fnt	fnt	-	-	-	0.293	fnt	80	fnt	fnt	fnt	0.270	fnt	74	fnt	fnt	fnt
C4_L1r_S1_S3_S4	fnt	fnt	-	-	-	0.289	fnt	79	fnt	fnt	fnt	0.273	fnt	74	fnt	fnt	fnt
depth-3/4 stacks, pooled (4)	fnt	fnt	-	-	-	0.304	fnt	83	fnt	fnt	fnt	0.304	fnt	83	fnt	fnt	fnt
C_L1r_X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C_X1_S1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C_S2_X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
unseen-containing stacks, pooled (3)	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C_L1r_X1m	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.234	fnt	64	fnt	fnt	fnt
C_S1_X1s	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C3_L1r_S1_X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt

Table 23. **Master grid, Gemma-3-12B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	fmt	fmt	fmt	fmt	fmt	0.520	fmt	fmt	0.507	fmt	fmt	0.502	fmt	fmt	0.506	fmt	fmt
L1b	fmt	fmt	fmt	fmt	fmt	0.441	fmt	fmt	0.475	fmt	fmt	0.488	fmt	fmt	0.402	fmt	fmt
L1r	fmt	fmt	fmt	fmt	fmt	0.459	fmt	fmt	0.463	fmt	fmt	0.472	fmt	fmt	0.447	fmt	fmt
L2	fmt	fmt	fmt	fmt	fmt	0.456	fmt	fmt	0.471	fmt	fmt	0.478	fmt	fmt	0.441	fmt	fmt
S1	fmt	fmt	fmt	fmt	fmt	0.456	fmt	fmt	0.457	fmt	fmt	0.483	fmt	fmt	0.452	fmt	fmt
S2	fmt	fmt	fmt	fmt	fmt	0.500	fmt	fmt	0.503	fmt	fmt	0.504	fmt	fmt	0.489	fmt	fmt
X1	0.152	fmt	-	-	-	0.310	+15.7	fmt	0.263	+11.0	fmt	0.289	+13.7	fmt	0.376	+22.3	fmt
C_L1b_S1	fmt	fmt	-	-	-	0.333	fmt	fmt	0.406	fmt	fmt	0.420	fmt	fmt	0.330	fmt	fmt
C_L1r_S1	fmt	fmt	-	-	-	0.336	fmt	fmt	0.389	fmt	fmt	0.416	fmt	fmt	0.344	fmt	fmt
C_S1_L1r	fmt	fmt	-	-	-	0.334	fmt	fmt	0.393	fmt	fmt	0.414	fmt	fmt	0.329	fmt	fmt
C_L2_S4	fmt	fmt	-	-	-	0.438	fmt	fmt	0.460	fmt	fmt	0.469	fmt	fmt	0.424	fmt	fmt
C_L1r_S3	fmt	fmt	-	-	-	0.438	fmt	fmt	0.462	fmt	fmt	0.469	fmt	fmt	0.423	fmt	fmt
C_S4_S3	fmt	fmt	-	-	-	0.491	fmt	fmt	0.506	fmt	fmt	0.503	fmt	fmt	0.481	fmt	fmt
<i>seen stacks, pooled (6)</i>	fmt	fmt	-	-	-	0.395	fmt	fmt	0.436	fmt	fmt	0.448	fmt	fmt	0.388	fmt	fmt
C3_L1r_S3_S4	fmt	fmt	-	-	-	0.419	fmt	fmt	0.454	fmt	fmt	0.467	fmt	fmt	0.417	fmt	fmt
C3_S1_S3_S4	fmt	fmt	-	-	-	0.425	fmt	fmt	0.452	fmt	fmt	0.465	fmt	fmt	0.431	fmt	fmt
C3_L1r_S1_S4	fmt	fmt	-	-	-	0.320	fmt	fmt	0.390	fmt	fmt	0.415	fmt	fmt	0.330	fmt	fmt
C4_L1r_S1_S3_S4	fmt	fmt	-	-	-	0.302	fmt	fmt	0.394	fmt	fmt	0.413	fmt	fmt	0.323	fmt	fmt
<i>depth-3/4 stacks, pooled (4)</i>	fmt	fmt	-	-	-	0.366	fmt	fmt	0.423	fmt	fmt	0.440	fmt	fmt	0.375	fmt	fmt
C_L1r_X1	0.102	fmt	-	-	-	0.195	+9.3	fmt	0.206	+10.4	fmt	0.224	+12.2	fmt	0.285	+18.3	fmt
C_X1_S1	0.172	fmt	-	-	-	0.317	+14.6	fmt	0.271	+10.0	fmt	0.296	+12.5	fmt	0.358	+18.6	fmt
C_S2_X1	0.201	fmt	-	-	-	0.368	+16.6	fmt	0.349	+14.8	fmt	0.364	+16.3	fmt	0.412	+21.1	fmt
<i>unseen-containing stacks, pooled (3)</i>	0.158	fmt	-	-	-	0.293	+13.5	fmt	0.275	+11.7	fmt	0.295	+13.7	fmt	0.352	+19.3	fmt
C_L1r_X1m	fmt	fmt	-	-	-	0.220	fmt	fmt	0.235	fmt	fmt	0.244	fmt	fmt	0.312	fmt	fmt
C_S1_X1s	0.255	fmt	-	-	-	0.395	+14.0	fmt	0.374	+12.0	fmt	0.395	+14.0	fmt	0.401	+14.7	fmt
C3_L1r_S1_X1	0.136	fmt	-	-	-	0.225	+8.9	fmt	0.236	+10.0	fmt	0.258	+12.2	fmt	0.260	+12.4	fmt
<i>Backward: input prediction, graded by execution</i>																	
L0	0.453	100	-	-	-	0.440	-1.3	97	0.447	-0.6	99	0.355	-9.8	78	0.469	+1.6	104
L1b	0.405	89	-	-	-	0.353	-5.2	78	0.427	+2.2	94	fmt	fmt	fmt	0.395	-1.0	87
L1r	0.420	93	-	-	-	0.394	-2.6	87	0.411	-0.9	91	fmt	fmt	fmt	0.428	+0.8	94
L2	0.403	89	-	-	-	0.378	-2.6	83	0.405	+0.2	89	fmt	fmt	fmt	0.431	+2.8	95
S1	0.424	94	-	-	-	0.375	-4.9	83	0.421	-0.2	93	fmt	fmt	fmt	0.461	+3.7	102
S2	0.443	98	-	-	-	0.444	+0.1	98	0.457	+1.4	101	fmt	fmt	fmt	0.454	+1.1	100
X1	0.333	74	-	-	-	0.305	-2.9	67	0.333	+0.0	74	fmt	fmt	fmt	0.393	+6.0	87
C_L1b_S1	0.349	77	-	-	-	0.317	-3.3	70	0.366	+1.7	81	fmt	fmt	fmt	0.370	+2.1	82
C_L1r_S1	0.356	79	-	-	-	0.319	-3.7	70	0.379	+2.2	84	fmt	fmt	fmt	0.397	+4.1	88
C_S1_L1r	0.337	74	-	-	-	0.319	-1.8	70	0.379	+4.1	84	fmt	fmt	fmt	0.407	+6.9	90
C_L2_S4	0.387	85	-	-	-	0.381	-0.6	84	0.416	+2.9	92	fmt	fmt	fmt	0.427	+4.0	94
C_L1r_S3	0.427	94	-	-	-	0.402	-2.6	89	0.423	-0.5	93	fmt	fmt	fmt	0.438	+1.1	97
C_S4_S3	0.434	96	-	-	-	0.438	+0.4	97	0.462	+2.8	102	fmt	fmt	fmt	0.469	+3.5	104
<i>seen stacks, pooled (6)</i>	0.382	84	-	-	-	0.363	-1.9	80	0.404	+2.2	89	fmt	fmt	fmt	0.418	+3.6	92
C3_L1r_S3_S4	0.398	88	-	-	-	0.386	-1.2	85	0.420	+2.2	93	fmt	fmt	fmt	0.447	+4.9	99
C3_S1_S3_S4	0.387	86	-	-	-	0.354	-3.3	78	0.419	+3.1	92	fmt	fmt	fmt	0.440	+5.2	97
C3_L1r_S1_S4	0.347	77	-	-	-	0.302	-4.5	67	0.376	+3.0	83	fmt	fmt	fmt	0.377	+3.0	83
C4_L1r_S1_S3_S4	0.329	73	-	-	-	0.293	-3.7	65	0.364	+3.5	80	fmt	fmt	fmt	0.399	+6.9	88
<i>depth-3/4 stacks, pooled (4)</i>	0.365	81	-	-	-	0.334	-3.2	74	0.395	+2.9	87	fmt	fmt	fmt	0.416	+5.0	92
C_L1r_X1	0.258	57	-	-	-	0.242	-1.6	53	0.279	+2.1	62	fmt	fmt	fmt	0.329	+7.2	73
C_X1_S1	0.348	77	-	-	-	0.282	-6.5	62	0.331	-1.7	73	fmt	fmt	fmt	0.410	+6.2	91
C_S2_X1	0.360	79	-	-	-	0.339	-2.0	75	0.357	-0.3	79	fmt	fmt	fmt	0.398	+3.8	88
<i>unseen-containing stacks, pooled (3)</i>	0.322	71	-	-	-	0.288	-3.4	64	0.322	+0.1	71	fmt	fmt	fmt	0.379	+5.7	84
C_L1r_X1m	0.273	60	-	-	-	0.269	-0.4	59	0.323	+5.0	71	fmt	fmt	fmt	0.349	+7.6	77
C_S1_X1s	0.412	91	-	-	-	0.371	-4.1	82	0.383	-3.0	84	fmt	fmt	fmt	0.448	+3.6	99
C3_L1r_S1_X1	0.279	62	-	-	-	0.226	-5.3	50	0.270	-0.9	60	fmt	fmt	fmt	0.337	+5.8	74

Table 24. **Master grid, CodeGemma-7B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.237	100	fmt	fmt	fmt	0.465	+22.9	197	0.439	+20.2	186	0.445	+20.8	188	0.462	+22.6	195
L1b	0.185	78	fmt	fmt	fmt	0.382	+19.7	161	0.425	+24.0	180	0.435	+25.1	184	0.345	+16.0	146
L1r	0.213	90	fmt	fmt	fmt	0.399	+18.6	169	0.417	+20.4	176	0.417	+20.4	176	0.390	+17.7	165
L2	0.208	88	fmt	fmt	fmt	0.412	+20.4	174	0.422	+21.4	178	0.422	+21.4	178	0.392	+18.3	166
S1	0.195	82	0.212	+1.7	90	0.405	+21.0	171	0.415	+22.1	176	0.421	+22.6	178	0.423	+22.8	179
S2	0.198	84	fmt	fmt	fmt	0.450	+25.2	190	0.429	+23.1	181	0.448	+25.0	189	0.444	+24.6	188
X1	0.143	61	-	-	-	0.271	+12.8	115	0.251	+10.8	106	0.247	+10.4	104	0.348	+20.4	147
C_L1b_S1	0.148	63	-	-	-	0.286	+13.7	121	0.372	+22.3	157	0.368	+21.9	156	0.287	+13.9	121
C_L1r_S1	0.144	61	-	-	-	0.315	+17.2	133	0.373	+22.9	158	0.367	+22.3	155	0.304	+16.0	129
C_S1_L1r	0.132	56	-	-	-	0.288	+15.6	122	0.363	+23.1	154	0.361	+22.8	153	0.299	+16.7	127
C_L2_S4	0.176	75	-	-	-	0.391	+21.5	165	0.423	+24.7	179	0.421	+24.5	178	0.374	+19.8	158
C_L1r_S3	0.183	77	-	-	-	0.396	+21.3	168	0.415	+23.2	175	0.410	+22.7	173	0.375	+19.2	159
C_S4_S3	0.202	85	-	-	-	0.448	+24.7	189	0.430	+22.9	182	0.441	+23.9	186	0.446	+24.5	189
seen stacks, pooled (6)	0.164	69	-	-	-	0.354	+19.0	150	0.396	+23.2	167	0.395	+23.0	167	0.348	+18.4	147
C3_L1r_S3_S4	0.157	66	-	-	-	0.376	+21.8	159	0.407	+25.0	172	0.411	+25.4	174	0.361	+20.3	152
C3_S1_S3_S4	0.158	67	-	-	-	0.367	+20.9	155	0.395	+23.7	167	0.405	+24.7	171	0.392	+23.4	166
C3_L1r_S1_S4	0.126	53	-	-	-	0.289	+16.3	122	0.367	+24.1	155	0.366	+23.9	155	0.301	+17.5	127
C4_L1r_S1_S3_S4	0.124	52	-	-	-	0.271	+14.7	114	0.361	+23.7	153	0.359	+23.5	152	0.285	+16.1	120
depth-3/4 stacks, pooled (4)	0.141	60	-	-	-	0.326	+18.4	138	0.383	+24.1	162	0.385	+24.4	163	0.335	+19.3	141
C_L1r_X1	0.118	50	-	-	-	0.194	+7.6	82	0.212	+9.4	90	0.197	+7.9	83	0.245	+12.8	104
C_X1_S1	0.144	61	-	-	-	0.285	+14.0	120	0.261	+11.6	110	0.247	+10.3	105	0.339	+19.4	143
C_S2_X1	0.188	80	-	-	-	0.344	+15.6	146	0.323	+13.5	136	0.326	+13.8	138	0.379	+19.1	160
unseen-containing stacks, pooled (3)	0.150	63	-	-	-	0.274	+12.4	116	0.265	+11.5	112	0.257	+10.7	109	0.321	+17.1	136
C_L1r_X1m	0.130	55	-	-	-	0.214	+8.4	90	0.256	+12.5	108	0.234	+10.4	99	0.288	+15.8	122
C_S1_X1s	0.186	79	-	-	-	0.338	+15.2	143	0.351	+16.5	148	0.339	+15.3	143	0.386	+20.0	163
C3_L1r_S1_X1	0.134	57	-	-	-	0.218	+8.4	92	0.241	+10.6	102	0.246	+11.2	104	0.254	+11.9	107
<i>Backward: input prediction, graded by execution</i>																	
L0	0.266	100	-	-	-	0.364	+9.8	137	0.324	+5.8	122	fmt	fmt	fmt	0.256	-1.0	96
L1b	0.268	101	-	-	-	0.309	+4.1	116	0.298	+3.1	112	fmt	fmt	fmt	0.193	-7.4	73
L1r	0.251	94	-	-	-	0.333	+8.3	125	0.301	+5.0	113	fmt	fmt	fmt	0.225	-2.6	85
L2	0.239	90	-	-	-	0.327	+8.7	123	0.291	+5.1	109	fmt	fmt	fmt	0.220	-1.9	83
S1	0.234	88	-	-	-	0.357	+12.3	134	0.337	+10.3	127	fmt	fmt	fmt	0.231	-0.3	87
S2	0.270	102	-	-	-	0.372	+10.2	140	0.351	+8.0	132	fmt	fmt	fmt	0.248	-2.3	93
X1	0.190	72	-	-	-	0.260	+7.0	98	0.217	+2.7	82	fmt	fmt	fmt	0.235	+4.4	88
C_L1b_S1	0.229	86	-	-	-	0.274	+4.5	103	0.276	+4.7	104	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_S1	0.170	64	-	-	-	0.289	+11.9	109	0.290	+12.0	109	fmt	fmt	fmt	0.208	+3.8	78
C_S1_L1r	0.168	63	-	-	-	0.284	+11.6	107	0.316	+14.8	119	fmt	fmt	fmt	0.203	+3.4	76
C_L2_S4	0.239	90	-	-	-	0.318	+8.0	120	0.299	+6.0	112	fmt	fmt	fmt	0.238	-0.1	90
C_L1r_S3	0.223	84	-	-	-	0.326	+10.4	123	0.314	+9.1	118	fmt	fmt	fmt	0.223	-0.0	84
C_S4_S3	0.257	97	-	-	-	0.372	+11.5	140	0.357	+10.0	134	fmt	fmt	fmt	0.242	-1.5	91
seen stacks, pooled (6)	0.214	81	-	-	-	0.310	+9.6	117	0.308	+9.4	116	fmt	fmt	fmt	fmt	fmt	fmt
C3_L1r_S3_S4	0.187	70	-	-	-	0.318	+13.1	120	0.312	+12.5	118	fmt	fmt	fmt	0.221	+3.4	83
C3_S1_S3_S4	0.231	87	-	-	-	0.338	+10.7	127	0.332	+10.0	125	fmt	fmt	fmt	0.234	+0.3	88
C3_L1r_S1_S4	0.173	65	-	-	-	0.274	+10.0	103	0.286	+11.3	108	fmt	fmt	fmt	0.191	+1.8	72
C4_L1r_S1_S3_S4	0.150	56	-	-	-	0.267	+11.7	101	0.290	+14.0	109	fmt	fmt	fmt	0.177	+2.7	67
depth-3/4 stacks, pooled (4)	0.185	70	-	-	-	0.299	+11.4	113	0.305	+12.0	115	fmt	fmt	fmt	0.206	+2.0	77
C_L1r_X1	fmt	fmt	-	-	-	0.208	fmt	78	0.174	fmt	66	fmt	fmt	fmt	0.178	fmt	67
C_X1_S1	0.202	76	-	-	-	0.303	+10.1	114	0.209	+0.6	79	fmt	fmt	fmt	0.227	+2.5	86
C_S2_X1	0.227	85	-	-	-	0.300	+7.3	113	0.241	+1.4	91	fmt	fmt	fmt	fmt	fmt	fmt
unseen-containing stacks, pooled (3)	fmt	fmt	-	-	-	0.271	fmt	102	0.208	fmt	78	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_X1m	fmt	fmt	-	-	-	0.231	fmt	87	0.211	fmt	79	fmt	fmt	fmt	0.205	fmt	77
C_S1_X1s	0.230	86	-	-	-	0.322	+9.3	121	0.230	+0.1	87	fmt	fmt	fmt	0.243	+1.4	92
C3_L1r_S1_X1	0.168	63	-	-	-	0.230	+6.2	87	fmt	fmt	fmt	fmt	fmt	fmt	0.191	+2.3	72

Table 25. **Master grid, Granite-3.1-8B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.285	100	0.293	+0.8	103	0.430	+14.5	151	0.416	+13.1	146	0.396	+11.1	139	0.413	+12.8	145
L1b	0.267	94	0.267	+0.0	94	0.338	+7.1	118	0.391	+12.5	137	0.381	+11.4	134	0.321	+5.4	113
L1r	0.268	94	0.260	-0.8	91	0.375	+10.8	132	0.383	+11.5	134	0.378	+11.1	133	0.376	+10.8	132
L2	0.251	88	0.248	-0.4	87	0.372	+12.0	130	0.387	+13.5	136	0.377	+12.6	132	0.366	+11.4	128
S1	0.254	89	0.253	-0.2	89	0.346	+9.1	121	0.379	+12.5	133	0.358	+10.4	126	0.350	+9.5	123
S2	0.272	96	0.276	+0.4	97	0.386	+11.4	136	0.415	+14.2	145	0.394	+12.1	138	0.398	+12.5	140
X1	0.147	51	-	-	-	0.216	+6.9	76	0.207	+6.0	73	0.188	+4.1	66	0.282	+13.5	99
C_L1b_S1	0.219	77	-	-	-	0.263	+4.4	92	0.342	+12.4	120	0.337	+11.8	118	0.261	+4.2	92
C_L1r_S1	0.204	72	-	-	-	0.279	+7.5	98	0.344	+14.0	121	0.338	+13.3	118	0.291	+8.7	102
C_S1_L1r	0.224	79	-	-	-	0.279	+5.4	98	0.326	+10.2	115	0.319	+9.5	112	0.291	+6.6	102
C_L2_S4	0.242	85	-	-	-	0.334	+9.2	117	0.376	+13.4	132	0.376	+13.4	132	0.341	+9.9	120
C_L1r_S3	0.259	91	-	-	-	0.350	+9.1	123	0.382	+12.3	134	0.376	+11.7	132	0.359	+10.1	126
C_S4_S3	0.270	95	-	-	-	0.380	+11.0	133	0.412	+14.2	144	0.389	+11.9	137	0.383	+11.3	134
<i>seen stacks, pooled (6)</i>	0.236	83	-	-	-	0.314	+7.8	110	0.364	+12.7	128	0.356	+12.0	125	0.321	+8.5	113
C3_L1r_S3_S4	0.257	90	-	-	-	0.333	+7.6	117	0.380	+12.3	133	0.375	+11.8	132	0.340	+8.3	119
C3_S1_S3_S4	0.242	85	-	-	-	0.316	+7.4	111	0.372	+13.1	131	0.358	+11.7	126	0.330	+8.9	116
C3_L1r_S1_S4	0.221	78	-	-	-	0.279	+5.7	98	0.333	+11.2	117	0.330	+10.9	116	0.283	+6.1	99
C4_L1r_S1_S3_S4	0.203	71	-	-	-	0.258	+5.5	90	0.325	+12.2	114	0.334	+13.1	117	0.270	+6.7	95
<i>depth-3/4 stacks, pooled (4)</i>	0.231	81	-	-	-	0.296	+6.6	104	0.352	+12.2	124	0.349	+11.9	123	0.306	+7.5	107
C_L1r_X1	0.136	48	-	-	-	0.157	+2.1	55	0.172	+3.6	60	0.163	+2.7	57	0.245	+11.0	86
C_X1_S1	0.153	54	-	-	-	0.222	+7.0	78	0.203	+5.0	71	0.186	+3.3	65	0.272	+12.0	96
C_S2_X1	0.209	73	-	-	-	0.263	+5.5	92	0.252	+4.4	89	0.240	+3.1	84	0.323	+11.4	113
<i>unseen-containing stacks, pooled (3)</i>	0.166	58	-	-	-	0.214	+4.8	75	0.209	+4.3	73	0.196	+3.1	69	0.280	+11.4	98
C_L1r_X1m	0.158	55	-	-	-	0.188	+3.0	66	0.204	+4.7	72	0.183	+2.6	64	0.279	+12.1	98
C_S1_X1s	0.206	72	-	-	-	0.298	+9.2	104	0.267	+6.1	94	0.269	+6.3	94	0.318	+11.2	111
C3_L1r_S1_X1	0.167	59	-	-	-	0.186	+1.8	65	0.195	+2.8	68	0.198	+3.1	70	0.238	+7.0	83
<i>Backward: input prediction, graded by execution</i>																	
L0	0.291	100	-	-	-	fmt	fmt	fmt	0.261	-3.0	90	fmt	fmt	fmt	0.272	-1.9	94
L1b	0.255	88	-	-	-	fmt	fmt	fmt	0.228	-2.7	79	fmt	fmt	fmt	0.211	-4.4	73
L1r	0.284	98	-	-	-	fmt	fmt	fmt	0.268	-1.7	92	0.150	-13.4	52	0.262	-2.3	90
L2	0.247	85	-	-	-	fmt	fmt	fmt	0.236	-1.1	81	fmt	fmt	fmt	0.222	-2.5	76
S1	0.305	105	-	-	-	fmt	fmt	fmt	0.220	-8.6	75	fmt	fmt	fmt	0.236	-6.9	81
S2	0.329	113	-	-	-	fmt	fmt	fmt	0.244	-8.5	84	fmt	fmt	fmt	0.266	-6.2	92
X1	fmt	fmt	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	0.207	fmt	71
C_L1b_S1	0.216	74	-	-	-	fmt	fmt	fmt	0.199	-1.8	68	fmt	fmt	fmt	0.194	-2.2	67
C_L1r_S1	0.284	98	-	-	-	0.206	-7.8	71	0.225	-5.9	77	fmt	fmt	fmt	0.234	-5.0	80
C_S1_L1r	0.262	90	-	-	-	0.208	-5.4	72	0.241	-2.2	83	fmt	fmt	fmt	0.242	-2.0	83
C_L2_S4	0.278	96	-	-	-	fmt	fmt	fmt	0.225	-5.3	78	fmt	fmt	fmt	0.213	-6.5	73
C_L1r_S3	0.296	102	-	-	-	fmt	fmt	fmt	0.240	-5.6	83	fmt	fmt	fmt	0.251	-4.5	86
C_S4_S3	0.334	115	-	-	-	fmt	fmt	fmt	0.254	-8.0	87	fmt	fmt	fmt	0.254	-8.0	87
<i>seen stacks, pooled (6)</i>	0.278	96	-	-	-	fmt	fmt	fmt	0.231	-4.8	79	fmt	fmt	fmt	0.231	-4.7	80
C3_L1r_S3_S4	0.331	114	-	-	-	fmt	fmt	fmt	0.238	-9.3	82	fmt	fmt	fmt	0.235	-9.6	81
C3_S1_S3_S4	0.323	111	-	-	-	fmt	fmt	fmt	0.214	-10.9	74	fmt	fmt	fmt	0.242	-8.1	83
C3_L1r_S1_S4	0.293	101	-	-	-	0.192	-10.0	66	0.223	-6.9	77	fmt	fmt	fmt	0.238	-5.5	82
C4_L1r_S1_S3_S4	0.305	105	-	-	-	fmt	fmt	fmt	0.216	-8.9	74	fmt	fmt	fmt	0.239	-6.5	82
<i>depth-3/4 stacks, pooled (4)</i>	0.313	108	-	-	-	fmt	fmt	fmt	0.223	-9.0	77	fmt	fmt	fmt	0.239	-7.4	82
C_L1r_X1	fmt	fmt	-	-	-	fmt	fmt	fmt	0.163	fmt	56	fmt	fmt	fmt	0.190	fmt	65
C_X1_S1	0.234	80	-	-	-	fmt	fmt	fmt	0.161	-7.3	55	fmt	fmt	fmt	0.242	+0.8	83
C_S2_X1	0.300	103	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	0.235	-6.5	81
<i>unseen-containing stacks, pooled (3)</i>	fmt	fmt	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	0.222	fmt	76
C_L1r_X1m	fmt	fmt	-	-	-	fmt	fmt	fmt	0.180	fmt	62	fmt	fmt	fmt	0.188	fmt	65
C_S1_X1s	0.282	97	-	-	-	fmt	fmt	fmt	0.173	-10.9	59	fmt	fmt	fmt	0.234	-4.8	81
C3_L1r_S1_X1	0.259	89	-	-	-	fmt	fmt	fmt	0.156	-10.3	54	fmt	fmt	fmt	0.226	-3.3	78

Table 26. **Routing, CodeLlama-7B.** MoLE mixtures of the eight per-transform experts on every condition and stack that has them: *router* = trained gate, *hard router* = its argmax, *uniform* = fixed uniform gate, *random* = gate frozen at random initialisation. Per system: raw accuracy, Δ points against the untuned model on the same items, and % of the untuned model’s clean-code accuracy. Every cell ran through the mixture engine.

Condition	untuned			router			hard router			uniform			random		
	base	acc	%	mole_router	acc	Δ	mole_hardrouter	acc	Δ	mole_uniform	acc	Δ	mole_random	acc	Δ
L0	0.258	100		0.429	+17.1	167	0.428	+17.0	167	0.428	+16.9	166	0.433	+17.5	169
L1b	0.196	76		0.381	+18.5	148	0.384	+18.8	149	0.371	+17.5	144	0.375	+17.9	146
L1r	0.205	80		0.383	+17.7	149	0.381	+17.6	148	0.390	+18.4	152	0.391	+18.6	152
L2	0.198	77		0.386	+18.8	150	0.384	+18.6	149	0.392	+19.3	152	0.393	+19.5	153
S1	0.167	65		0.395	+22.8	154	0.393	+22.6	153	0.395	+22.8	154	0.394	+22.7	153
S2	0.191	74		0.421	+23.0	164	0.421	+22.9	164	0.415	+22.3	161	0.411	+22.0	160
X1	0.118	46		0.279	+16.1	109	0.283	+16.5	110	0.283	+16.6	110	0.283	+16.5	110
C_L1b_S1	0.141	55		0.325	+18.4	126	0.324	+18.3	126	0.310	+16.8	121	0.312	+17.1	121
C_L1r_S1	0.131	51		0.346	+21.5	135	0.339	+20.8	132	0.318	+18.7	124	0.320	+18.9	125
C_S1_L1r	0.135	53		0.318	+18.4	124	0.324	+18.9	126	0.302	+16.8	118	0.303	+16.8	118
C_L2_S4	0.176	68		0.383	+20.8	149	0.379	+20.3	148	0.380	+20.4	148	0.379	+20.3	147
C_L1r_S3	0.172	67		0.391	+21.9	152	0.389	+21.7	151	0.383	+21.1	149	0.380	+20.8	148
C_S4_S3	0.185	72		0.415	+22.9	161	0.413	+22.7	161	0.409	+22.4	159	0.409	+22.4	159
C3_L1r_S3_S4	0.150	58		0.375	+22.5	146	0.376	+22.6	146	0.365	+21.5	142	0.363	+21.3	141
C3_S1_S3_S4	0.128	50		0.377	+24.8	147	0.378	+24.9	147	0.367	+23.8	143	0.367	+23.9	143
C3_L1r_S1_S4	0.126	49		0.325	+19.9	126	0.326	+20.0	127	0.306	+18.0	119	0.305	+17.9	119
C4_L1r_S1_S3_S4	0.089	35		0.334	+24.5	130	0.336	+24.7	131	0.296	+20.7	115	0.298	+20.9	116
C_L1r_X1	0.102	40		0.189	+8.7	74	0.186	+8.4	72	0.194	+9.2	76	0.190	+8.8	74
C_X1_S1	0.104	40		0.291	+18.7	113	0.286	+18.2	111	0.293	+18.9	114	0.295	+19.1	115
C_S2_X1	0.145	56		0.326	+18.1	127	0.328	+18.3	128	0.329	+18.4	128	0.329	+18.4	128
C_L1r_X1m	0.114	44		0.227	+11.3	88	0.224	+11.0	87	0.221	+10.7	86	0.221	+10.6	86
C_S1_X1s	0.148	58		0.350	+20.2	136	0.348	+20.0	135	0.344	+19.6	134	0.343	+19.5	134
C3_L1r_S1_X1	0.102	40		0.242	+14.0	94	0.245	+14.3	95	0.234	+13.2	91	0.239	+13.8	93

Table 27. **Merging, CodeLlama-7B.** Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned			TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES		
	base	acc	%	merge_ties	acc	Δ	merge_dare_ties	acc	Δ	merge_dare_linear	acc	Δ	l0merge_ties	acc	Δ	l0merge_dare_ties	acc	Δ
L0	0.257	100		0.366	+11.0	143	0.426	+16.9	166	fnt fnt	fnt		0.394	+13.7	153	0.413	+15.6	161
L1b	0.197	77		0.299	+10.1	116	0.373	+17.6	145	fnt fnt	fnt		0.331	+13.4	129	0.349	+15.1	136
L1r	0.207	81		0.323	+11.6	126	0.381	+17.4	148	fnt fnt	fnt		0.338	+13.1	132	0.352	+14.5	137
L2	0.202	79		0.317	+11.5	123	0.375	+17.4	146	fnt fnt	fnt		0.337	+13.5	131	0.355	+15.3	138
S1	0.168	66		0.322	+15.4	125	0.376	+20.8	146	fnt fnt	fnt		0.342	+17.3	133	0.365	+19.6	142
S2	0.193	75		0.340	+14.7	132	0.392	+19.9	153	fnt fnt	fnt		0.365	+17.2	142	0.379	+18.5	147
X1	0.119	46		0.234	+11.4	91	0.276	+15.7	107	fnt fnt	fnt		-	-	-	-	-	-
C_L1b_S1	0.141	55		0.245	+10.4	95	0.302	+16.0	117	fnt fnt	fnt		0.269	+12.8	105	0.280	+13.9	109
C_L1r_S1	0.131	51		0.255	+12.4	99	0.301	+17.0	117	fnt fnt	fnt		0.277	+14.6	108	0.283	+15.2	110
C_S1_L1r	0.135	53		0.245	+11.0	95	0.301	+16.6	117	fnt fnt	fnt		0.273	+13.8	106	0.282	+14.7	110
C_L2_S4	0.176	68		0.299	+12.3	116	0.363	+18.7	141	fnt fnt	fnt		0.325	+14.9	126	0.349	+17.3	136
C_L1r_S3	0.172	67		0.307	+13.5	120	0.368	+19.6	143	fnt fnt	fnt		0.326	+15.4	127	0.338	+16.6	131
C_S4_S3	0.186	72		0.341	+15.5	133	0.398	+21.2	155	fnt fnt	fnt		0.367	+18.1	143	0.385	+19.9	150
C3_L1r_S3_S4	0.150	58		0.298	+14.8	116	0.349	+19.9	136	fnt fnt	fnt		0.313	+16.3	122	0.328	+17.8	128
C3_S1_S3_S4	0.128	50		0.271	+14.3	106	0.340	+21.1	132	fnt fnt	fnt		0.295	+16.6	115	0.315	+18.7	123
C3_L1r_S1_S4	0.126	49		0.240	+11.4	94	0.296	+17.0	115	fnt fnt	fnt		0.272	+14.6	106	0.283	+15.7	110
C4_L1r_S1_S3_S4	0.089	35		0.217	+12.8	85	0.290	+20.0	113	fnt fnt	fnt		0.243	+15.4	95	0.256	+16.7	100
C_L1r_X1	0.102	40		0.171	+6.9	67	0.193	+9.1	75	fnt fnt	fnt		-	-	-	-	-	-
C_X1_S1	0.104	40		0.223	+12.0	87	0.290	+18.6	113	fnt fnt	fnt		-	-	-	-	-	-
C_S2_X1	0.145	57		0.265	+12.0	103	0.317	+17.1	123	fnt fnt	fnt		-	-	-	-	-	-
C_L1r_X1m	0.112	44		0.190	+7.8	74	0.215	+10.3	84	fnt fnt	fnt		-	-	-	-	-	-
C_S1_X1s	0.151	59		0.301	+15.0	117	0.338	+18.8	132	fnt fnt	fnt		-	-	-	-	-	-
C3_L1r_S1_X1	0.104	40		0.191	+8.7	74	0.232	+12.8	90	fnt fnt	fnt		-	-	-	-	-	-

Table 28. **Merging, Granite-3.1-8B.** Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned base		TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES		
	acc	%	merge_ties	acc	Δ	merge_dare_ties	acc	Δ	merge_dare_linear	acc	Δ	l0merge_ties	acc	Δ	l0merge_dare_ties	acc	Δ
L0	0.285	100	0.383	+9.8	134	0.433	+14.8	152	fmt	fmt	fmt	-	-	-	-	-	-
L1b	0.266	93	0.315	+4.9	110	0.350	+8.4	123	fmt	fmt	fmt	-	-	-	-	-	-
L1r	0.268	94	0.353	+8.5	124	0.384	+11.6	135	fmt	fmt	fmt	-	-	-	-	-	-
L2	0.251	88	0.342	+9.0	120	0.378	+12.6	133	fmt	fmt	fmt	-	-	-	-	-	-
S1	0.254	89	0.323	+6.9	113	0.371	+11.7	130	fmt	fmt	fmt	-	-	-	-	-	-
S2	0.272	96	0.358	+8.5	125	0.402	+13.0	141	fmt	fmt	fmt	-	-	-	-	-	-
X1	0.147	51	0.199	+5.3	70	0.231	+8.4	81	fmt	fmt	fmt	-	-	-	-	-	-
C_L1b_S1	0.219	77	0.247	+2.9	87	0.279	+6.0	98	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_S1	0.204	72	0.266	+6.1	93	0.301	+9.7	106	fmt	fmt	fmt	-	-	-	-	-	-
C_S1_L1r	0.224	79	0.262	+3.8	92	0.293	+6.9	103	fmt	fmt	fmt	-	-	-	-	-	-
C_L2_S4	0.242	85	0.319	+7.7	112	0.351	+10.9	123	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_S3	0.259	91	0.324	+6.5	114	0.365	+10.6	128	fmt	fmt	fmt	-	-	-	-	-	-
C_S4_S3	0.270	95	0.353	+8.3	124	0.394	+12.4	138	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S3_S4	0.257	90	0.307	+5.0	108	0.348	+9.1	122	fmt	fmt	fmt	-	-	-	-	-	-
C3_S1_S3_S4	0.242	85	0.315	+7.3	111	0.342	+10.1	120	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S1_S4	0.221	78	0.271	+4.9	95	0.287	+6.6	101	fmt	fmt	fmt	-	-	-	-	-	-
C4_L1r_S1_S3_S4	0.203	71	0.251	+4.8	88	0.281	+7.8	99	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_X1	0.136	48	0.153	+1.7	54	0.173	+3.7	61	fmt	fmt	fmt	-	-	-	-	-	-
C_X1_S1	0.153	54	0.200	+4.7	70	0.244	+9.1	86	fmt	fmt	fmt	-	-	-	-	-	-
C_S2_X1	0.209	73	0.246	+3.7	86	0.272	+6.3	95	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_X1m	0.158	55	0.173	+1.5	61	0.211	+5.3	74	fmt	fmt	fmt	-	-	-	-	-	-
C_S1_X1s	0.206	72	0.259	+5.3	91	0.293	+8.7	103	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S1_X1	0.167	59	0.204	+3.7	72	0.217	+5.0	76	fmt	fmt	fmt	-	-	-	-	-	-

Table 29. **Merging, Llama-3.1-8B.** Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned base		TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES		
	acc	%	merge_ties	acc	Δ	merge_dare_ties	acc	Δ	merge_dare_linear	acc	Δ	l0merge_ties	acc	Δ	l0merge_dare_ties	acc	Δ
L0	0.259	100	0.398	+14.0	154	0.462	+20.3	179	0.292	+3.4	113	-	-	-	-	-	-
L1b	0.222	86	0.336	+11.4	130	0.386	+16.4	150	0.253	+3.1	98	-	-	-	-	-	-
L1r	0.242	94	0.348	+10.6	135	0.398	+15.6	154	0.234	-0.8	91	-	-	-	-	-	-
L2	0.215	83	0.353	+13.8	137	0.396	+18.1	153	0.246	+3.1	95	-	-	-	-	-	-
S1	0.213	83	0.335	+12.2	130	0.412	+19.9	160	0.256	+4.3	99	-	-	-	-	-	-
S2	0.208	80	0.361	+15.3	140	0.437	+23.0	169	0.256	+4.9	99	-	-	-	-	-	-
X1	0.132	51	0.217	+8.5	84	0.266	+13.4	103	fmt	fmt	fmt	-	-	-	-	-	-
C_L1b_S1	0.152	59	0.252	+10.1	98	0.307	+15.6	119	0.211	+6.0	82	-	-	-	-	-	-
C_L1r_S1	0.162	63	0.273	+11.1	106	0.328	+16.6	127	0.206	+4.4	80	-	-	-	-	-	-
C_S1_L1r	0.159	62	0.247	+8.9	96	0.306	+14.8	119	0.200	+4.2	78	-	-	-	-	-	-
C_L2_S4	0.206	80	0.311	+10.5	120	0.383	+17.8	149	0.226	+2.0	88	-	-	-	-	-	-
C_L1r_S3	0.198	77	0.334	+13.5	129	0.387	+18.9	150	0.222	+2.3	86	-	-	-	-	-	-
C_S4_S3	0.214	83	0.357	+14.3	138	0.437	+22.3	169	0.255	+4.1	99	-	-	-	-	-	-
C3_L1r_S3_S4	0.184	71	0.306	+12.2	119	0.370	+18.6	143	0.199	+1.6	77	-	-	-	-	-	-
C3_S1_S3_S4	0.157	61	0.308	+15.1	119	0.377	+22.0	146	0.220	+6.2	85	-	-	-	-	-	-
C3_L1r_S1_S4	0.166	64	0.260	+9.4	101	0.319	+15.3	124	0.183	+1.7	71	-	-	-	-	-	-
C4_L1r_S1_S3_S4	0.134	52	0.235	+10.1	91	0.310	+17.6	120	0.179	+4.5	69	-	-	-	-	-	-
C_L1r_X1	0.100	39	0.147	+4.6	57	0.184	+8.3	71	fmt	fmt	fmt	-	-	-	-	-	-
C_X1_S1	0.130	50	0.233	+10.3	90	0.296	+16.6	115	fmt	fmt	fmt	-	-	-	-	-	-
C_S2_X1	0.169	66	0.263	+9.4	102	0.317	+14.8	123	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_X1m	0.114	44	0.162	+4.8	63	0.217	+10.3	84	0.126	+1.2	49	-	-	-	-	-	-
C_S1_X1s	0.198	77	0.287	+8.9	111	0.349	+15.1	135	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S1_X1	0.105	41	0.170	+6.6	66	0.215	+11.0	83	fmt	fmt	fmt	-	-	-	-	-	-

Table 30. **Merging, StarCoder2-15B**. Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned base		TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES			
			merge_ties			merge_dare_ties			merge_dare_linear			l0merge_ties			l0merge_dare_ties			
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	
L0	fmt	fmt	0.484	fmt	fmt	0.539	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
L1b	fmt	fmt	0.366	fmt	fmt	0.426	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
L1r	fmt	fmt	0.414	fmt	fmt	0.476	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
L2	fmt	fmt	0.401	fmt	fmt	0.466	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
S1	fmt	fmt	0.451	fmt	fmt	0.487	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
S2	fmt	fmt	0.440	fmt	fmt	0.518	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
X1	fmt	fmt	0.283	fmt	fmt	0.343	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C_L1b_S1	fmt	fmt	0.321	fmt	fmt	0.354	fmt	fmt	0.063	fmt	fmt	fmt	–	–	–	–	–	–
C_L1r_S1	fmt	fmt	0.344	fmt	fmt	0.389	fmt	fmt	0.047	fmt	fmt	fmt	–	–	–	–	–	–
C_S1_L1r	fmt	fmt	0.346	fmt	fmt	0.387	fmt	fmt	0.041	fmt	fmt	fmt	–	–	–	–	–	–
C_L2_S4	fmt	fmt	0.381	fmt	fmt	0.438	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C_L1r_S3	fmt	fmt	0.390	fmt	fmt	0.459	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C_S4_S3	fmt	fmt	0.442	fmt	fmt	0.506	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C3_L1r_S3_S4	fmt	fmt	0.377	fmt	fmt	0.436	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C3_S1_S3_S4	fmt	fmt	0.393	fmt	fmt	0.464	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C3_L1r_S1_S4	fmt	fmt	0.332	fmt	fmt	0.376	fmt	fmt	0.045	fmt	fmt	fmt	–	–	–	–	–	–
C4_L1r_S1_S3_S4	fmt	fmt	0.301	fmt	fmt	0.358	fmt	fmt	0.026	fmt	fmt	fmt	–	–	–	–	–	–
C_L1r_X1	fmt	fmt	0.215	fmt	fmt	0.259	fmt	fmt	0.058	fmt	fmt	fmt	–	–	–	–	–	–
C_X1_S1	fmt	fmt	0.295	fmt	fmt	0.371	fmt	fmt	0.067	fmt	fmt	fmt	–	–	–	–	–	–
C_S2_X1	fmt	fmt	0.341	fmt	fmt	0.403	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C_L1r_X1m	fmt	fmt	0.252	fmt	fmt	0.317	fmt	fmt	0.065	fmt	fmt	fmt	–	–	–	–	–	–
C_S1_X1s	fmt	fmt	0.378	fmt	fmt	0.427	fmt	fmt	fmt	fmt	fmt	fmt	–	–	–	–	–	–
C3_L1r_S1_X1	fmt	fmt	0.241	fmt	fmt	0.286	fmt	fmt	0.058	fmt	fmt	fmt	–	–	–	–	–	–

Table 31. **RQ1: routing and merging on stacks of seen transforms**. Paired differences in points on the ten stacks (six depth-2, four depth-3/4), CodeLlama-7B, 394 programs common to every cell; the last column repeats the contrasts on the four stacks that contain the unseen family (319 programs; merges other than DARE-TIES have no cells there). Routing is worth about a point over a random gate and nothing over breadth; the mixture itself carries the gain (mole_uniform – base); a hard argmax router is indistinguishable from the soft one. The best merge (DARE-TIES) matches the clean-code control and the other four fall below it. Bold intervals exclude zero. Every arm ran through the mixture engine (obtune.mole.eval_mole); an earlier run through vLLM had written the untuned model under the routing arms’ names and was withdrawn.

Contrast	all ten seen	depth 2	depth 3–4	unseen inside
<i>Routing</i>				
mole_router – mole_random	+1.34 [+0.48, +2.17]	+1.11 [+0.28, +1.93]	+1.67 [+0.47, +2.86]	–0.03 [–0.99, +0.84]
mole_router – mole_uniform	+1.40 [+0.52, +2.25]	+1.17 [+0.31, +2.02]	+1.74 [+0.53, +2.90]	+0.10 [–0.84, +0.99]
mole_hardrouter – mole_router	+0.03 [–0.13, +0.19]	–0.03 [–0.24, +0.18]	+0.13 [–0.11, +0.38]	–0.13 [–0.44, +0.18]
mole_router – mono_all	–0.28 [–1.57, +1.09]	–0.06 [–1.38, +1.30]	–0.61 [–2.10, +0.91]	+2.88 [+1.33, +4.41]
mole_uniform – base	+19.79 [+17.40, +22.23]	+19.16 [+16.77, +21.58]	+20.72 [+18.16, +23.31]	+14.98 [+12.81, +17.29]
<i>Merging</i>				
merge_dare_ties – tuned_L0	+0.52 [–0.15, +1.20]	+0.61 [–0.16, +1.41]	+0.38 [–0.47, +1.25]	+0.86 [–0.05, +1.78]
merge_ties – tuned_L0	–5.50 [–6.86, –4.13]	–5.19 [–6.59, –3.77]	–5.95 [–7.47, –4.44]	–
merge_dare_linear – tuned_L0	–20.56 [–22.88, –18.25]	–20.60 [–23.00, –18.22]	–20.49 [–22.86, –18.15]	–
l0merge_ties – tuned_L0	–3.14 [–4.21, –2.12]	–2.81 [–3.98, –1.67]	–3.64 [–4.91, –2.41]	–
l0merge_dare_ties – tuned_L0	–1.74 [–2.57, –0.91]	–1.40 [–2.30, –0.47]	–2.24 [–3.26, –1.23]	–
<i>Breadth, for reference</i>				
mono_all – tuned_L0	+3.67 [+2.02, +5.24]	+3.34 [+1.61, +5.04]	+4.15 [+2.35, +5.91]	–1.49 [–3.08, +0.08]

Table 32. **Weight-space merging on the panel.** The best merge (DARE-TIES of the clean adapter and the five specialists) against the clean-code control, breadth and the anchored objective, pooled over the six seen stacks and the four stacks containing the unseen family; breadth and anchoring against the control in the last two columns. Points, paired on the programs common to every cell, program-clustered bootstrap. Bold intervals exclude zero. Pre-registered per model: merging does not beat the control on seen stacks (CONFIRMED iff $ci_{hi} \leq +1$) and stays below breadth ($ci_{hi} < 0$).

Stacks	merge-clean	merge-mono	merge-cons	mono-clean	cons-clean
CodeLlama-7B					
seen (6)	+0.83 [+0.01, +1.63]	-2.80 [-4.43, -1.19]	-4.01 [-5.29, -2.70]	+3.63 [+1.96, +5.27]	+4.84 [+3.49, +6.21]
unseen inside (4)	+0.86 [-0.05, +1.78]	+2.35 [+0.63, +3.97]	-1.39 [-2.72, -0.05]	-1.49 [-3.08, +0.08]	+2.25 [+0.86, +3.61]
CodeLlama-13B					
seen (6)	+1.67 [+0.83, +2.51]	-1.40 [-2.98, +0.12]	-2.41 [-3.85, -1.05]	+3.07 [+1.55, +4.63]	+4.08 [+2.72, +5.40]
unseen inside (4)	+1.75 [+0.78, +2.80]	+2.48 [+0.89, +4.08]	+0.08 [-1.23, +1.39]	-0.73 [-2.23, +0.76]	+1.67 [+0.45, +2.82]
Llama-3.1-8B					
seen (6)	+2.31 [+1.27, +3.38]	-0.85 [-2.41, +0.72]	-2.80 [-4.26, -1.35]	+3.16 [+1.52, +4.84]	+5.11 [+3.63, +6.58]
unseen inside (4)	+2.56 [+1.49, +3.61]	+3.09 [+1.54, +4.60]	+0.81 [-0.47, +2.12]	-0.52 [-2.07, +1.02]	+1.75 [+0.39, +3.11]
StarCoder2-15B					
seen (6)	-0.69 [-1.57, +0.13]	-3.27 [-4.67, -1.86]	-5.73 [-7.05, -4.47]	+2.57 [+1.28, +3.86]	+5.04 [+3.82, +6.26]
unseen inside (4)	+1.15 [+0.08, +2.17]	+1.70 [+0.05, +3.32]	+0.39 [-0.92, +1.80]	-0.55 [-2.12, +1.02]	+0.76 [-0.47, +2.02]
Gemma-3-12B					
seen (6)	+1.27 [+0.32, +2.22]	-2.61 [-4.13, -1.11]	-3.91 [-5.28, -2.52]	+3.88 [+2.18, +5.64]	+5.17 [+3.69, +6.64]
unseen inside (4)	+0.42 [-0.55, +1.36]	+1.83 [+0.39, +3.40]	-0.18 [-1.57, +1.20]	-1.41 [-3.01, +0.13]	+0.60 [-0.78, +1.96]
CodeGemma-7B					
seen (6)	+0.44 [-0.48, +1.36]	-3.83 [-5.34, -2.32]	-3.59 [-5.08, -2.24]	+4.27 [+2.83, +5.63]	+4.03 [+2.67, +5.46]
unseen inside (4)	-0.10 [-1.02, +0.81]	-0.13 [-1.67, +1.44]	+0.92 [-0.47, +2.35]	+0.03 [-1.55, +1.57]	-1.02 [-2.38, +0.29]
Granite-3.1-8B					
seen (6)	+1.57 [+0.76, +2.46]	-3.23 [-4.92, -1.52]	-2.32 [-4.01, -0.61]	+4.80 [+3.05, +6.53]	+3.89 [+2.12, +5.67]
unseen inside (4)	+2.09 [+1.10, +3.09]	+2.20 [+0.50, +3.84]	+2.98 [+1.28, +4.65]	-0.10 [-1.78, +1.62]	-0.89 [-2.59, +0.89]

Table 33. **The direction ratio: how much of a forward gain survives the task being run backwards.** Forward is output prediction; backward is input prediction on the same programs, graded by execution. No adapter is trained on the backward task, so any backward gain is transferred program understanding rather than task fitting. $DR < 0$ means the arm bought forward accuracy by *losing* backward competence. Every adaptation method that fits the forward task directly is negative; only anchoring and family exposure are not. Both directions use the same 412 programs and the same six conditions. Bold intervals exclude zero; the ratio is descriptive and is read with its components.

Method	Arm	Forward gain	Backward gain	DR
Format-only control	formatonly	+1.47 [+0.90, +2.10]	-0.44 [-0.81, -0.08]	-0.30
Clean-code tuning	tuned_L0	+19.34 [+16.87, +21.81]	-1.60 [-3.41, +0.35]	-0.08
Training breadth	mono_all	+19.46 [+16.68, +22.01]	-0.89 [-2.86, +1.04]	-0.05
Paired consistency (ours)	cons_lam3	+20.88 [+18.07, +23.49]	+0.62 [-1.56, +2.81]	+0.03
Family exposure	tuned_X1	+17.35 [+14.98, +19.73]	+1.63 [+0.11, +3.20]	+0.09

Table 34. **RQ4: the divergence result across the panel**, on stacks containing the unseen family. cons-mono is anchoring’s advantage over breadth; fam is an arm trained on the unseen family’s trainable sibling, included because family exposure is the largest effect in the table and replicates on every model. Bold intervals exclude zero.

Model	Lineage	cons-mono	cons-clean	fam-clean
CodeLlama-7B	Meta	+3.84 [+1.90, +5.70]	+2.13 [+0.66, +3.60]	+5.93 [+4.10, +7.78]
CodeLlama-13B	Meta	+3.22 [+1.74, +4.65]	+2.02 [+0.58, +3.53]	+5.97 [+4.11, +7.91]
CodeLlama-34B	Meta	+3.95 [+2.13, +5.81]	+3.10 [+1.63, +4.58]	+4.19 [+2.48, +5.94]
Llama-3.1-8B	Meta	+2.87 [+1.20, +4.62]	+1.86 [+0.27, +3.53]	+6.67 [+4.58, +8.71]
StarCoder2-15B	BigCode	+1.36 [−0.08, +2.71]	+0.35 [−1.13, +1.86]	+3.64 [+1.90, +5.35]
Gemma-3-12B	Google	+2.44 [+0.97, +3.99]	−0.08 [−1.67, +1.51]	+5.35 [+3.21, +7.37]
CodeGemma-7B	Google	−1.59 [−3.26, +0.12]	−2.40 [−4.11, −0.85]	+5.23 [+3.25, +7.21]
Granite-3.1-8B	IBM	−1.47 [−2.99, +0.00]	−1.86 [−3.80, +0.12]	+7.29 [+5.19, +9.65]

Table 35. **Every intervention against every kind of obfuscation, across the eight-model panel**. For each system and direction, Δ is the mean change in points against the untuned model and % the mean accuracy as a percentage of the untuned model’s clean-code accuracy, over the cells that exist and pass the format gate (in parentheses: the number of cells; after the semicolon, the number gated as *fnt*). Systems: ICL = base_1shot, clean LoRA = tuned_L0, breadth = mono_all, anchored = cons_lam3, family = tuned_X1. *seen* = L1b, L1r, L2, S1, S2; *seen stacks* = the six depth-2 and four depth-3/4 stacks of seen transforms; *unseen stacks* = the three depth-2 and one depth-3 stacks with the X1 family inside. Backward cells survive the gate almost only on CodeLlama-7B, whose grid (Table 18) is the one fully interpretable backward read; the per-model grids for the whole panel are in Appendix B.

System	measure	clean	seen	unseen family	seen stacks	unseen stacks
ICL	fwd Δ pts	+3.8 (3; 5 fnt)	+3.2 (16; 24 fnt)	−	−	−
	fwd % of clean	115% (3; 5 fnt)	97% (16; 24 fnt)	−	−	−
clean LoRA	fwd Δ pts	+19.0 (5; 3 fnt)	+18.5 (30; 10 fnt)	+13.7 (7; 1 fnt)	+16.7 (60; 20 fnt)	+11.5 (28; 4 fnt)
	fwd % of clean	175% (5; 3 fnt)	152% (25; 15 fnt)	102% (5; 3 fnt)	129% (50; 30 fnt)	95% (20; 12 fnt)
	bwd Δ pts	−0.4 (7; 1 fnt)	−0.3 (28; 12 fnt)	+0.2 (5; 3 fnt)	+0.8 (54; 26 fnt)	−0.8 (20; 12 fnt)
	bwd % of clean	101% (7; 1 fnt)	97% (32; 8 fnt)	79% (5; 3 fnt)	90% (63; 17 fnt)	79% (21; 11 fnt)
breadth	fwd Δ pts	+17.0 (5; 3 fnt)	+19.7 (30; 10 fnt)	+10.7 (7; 1 fnt)	+20.9 (60; 20 fnt)	+10.9 (28; 4 fnt)
	fwd % of clean	167% (5; 3 fnt)	157% (25; 15 fnt)	91% (5; 3 fnt)	146% (50; 30 fnt)	93% (20; 12 fnt)
	bwd Δ pts	−2.2 (6; 2 fnt)	−1.6 (26; 14 fnt)	−1.1 (4; 4 fnt)	−0.0 (50; 30 fnt)	−4.3 (15; 17 fnt)
	bwd % of clean	95% (6; 2 fnt)	91% (29; 11 fnt)	73% (4; 4 fnt)	88% (52; 28 fnt)	64% (17; 15 fnt)
anchored	fwd Δ pts	+17.6 (5; 3 fnt)	+21.0 (30; 10 fnt)	+13.1 (7; 1 fnt)	+22.0 (60; 20 fnt)	+12.8 (28; 4 fnt)
	fwd % of clean	169% (5; 3 fnt)	161% (25; 15 fnt)	99% (5; 3 fnt)	148% (50; 30 fnt)	98% (20; 12 fnt)
	bwd Δ pts	−4.4 (6; 2 fnt)	−5.1 (22; 18 fnt)	−2.6 (4; 4 fnt)	−4.2 (39; 41 fnt)	−3.4 (9; 23 fnt)
	bwd % of clean	89% (6; 2 fnt)	86% (26; 14 fnt)	75% (5; 3 fnt)	81% (49; 31 fnt)	75% (9; 23 fnt)
family	fwd Δ pts	+18.5 (5; 3 fnt)	+17.8 (30; 10 fnt)	+20.1 (7; 1 fnt)	+17.5 (60; 20 fnt)	+16.8 (28; 4 fnt)
	fwd % of clean	173% (5; 3 fnt)	150% (25; 15 fnt)	128% (5; 3 fnt)	132% (50; 30 fnt)	116% (20; 12 fnt)
	bwd Δ pts	−1.4 (7; 1 fnt)	−2.8 (31; 9 fnt)	+1.6 (5; 3 fnt)	−1.0 (58; 22 fnt)	−0.9 (21; 11 fnt)
	bwd % of clean	96% (7; 1 fnt)	87% (33; 7 fnt)	80% (6; 2 fnt)	84% (59; 21 fnt)	76% (23; 9 fnt)